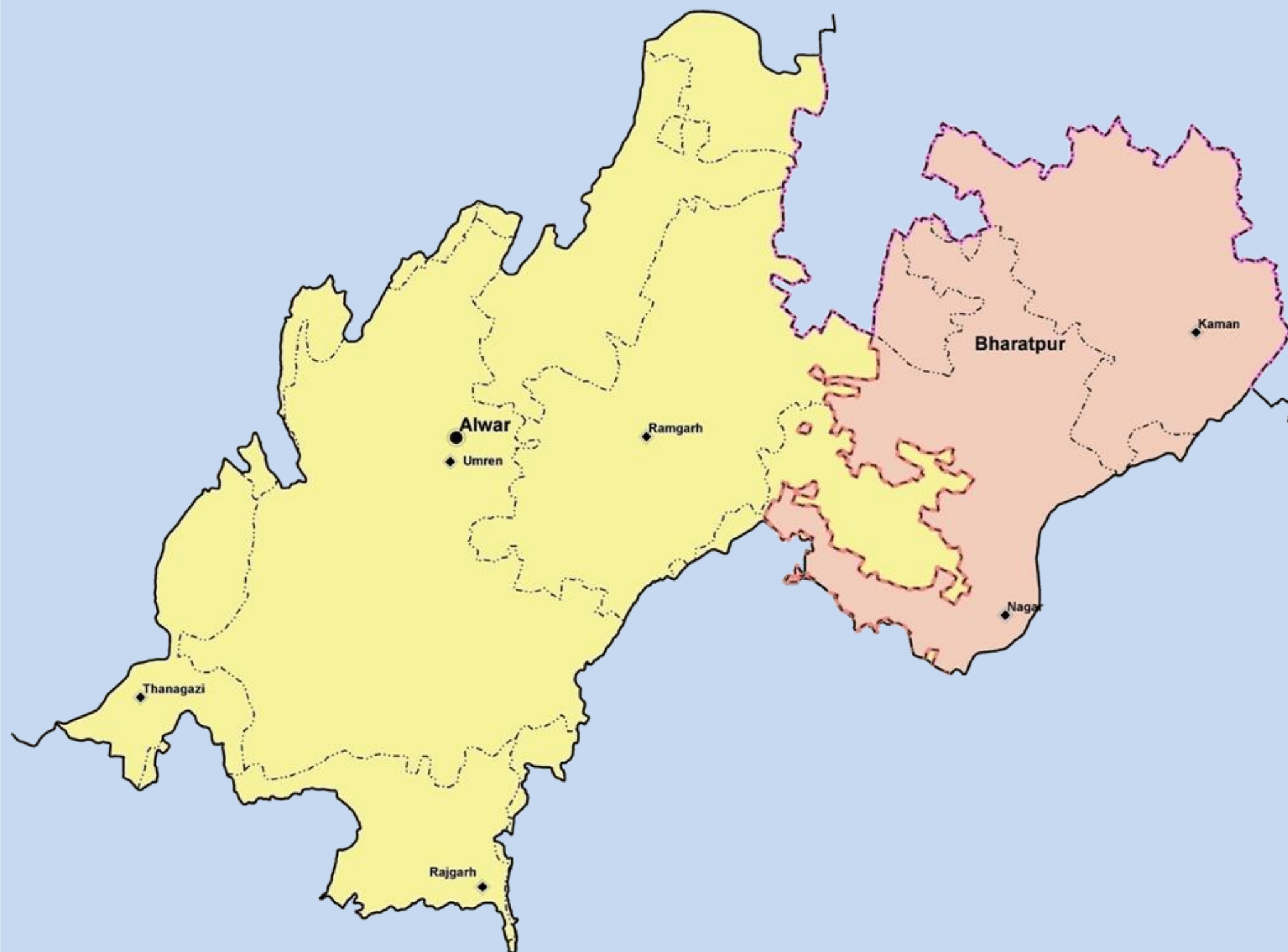


Hydrogeological Atlas of Rajasthan

Ruparail River Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

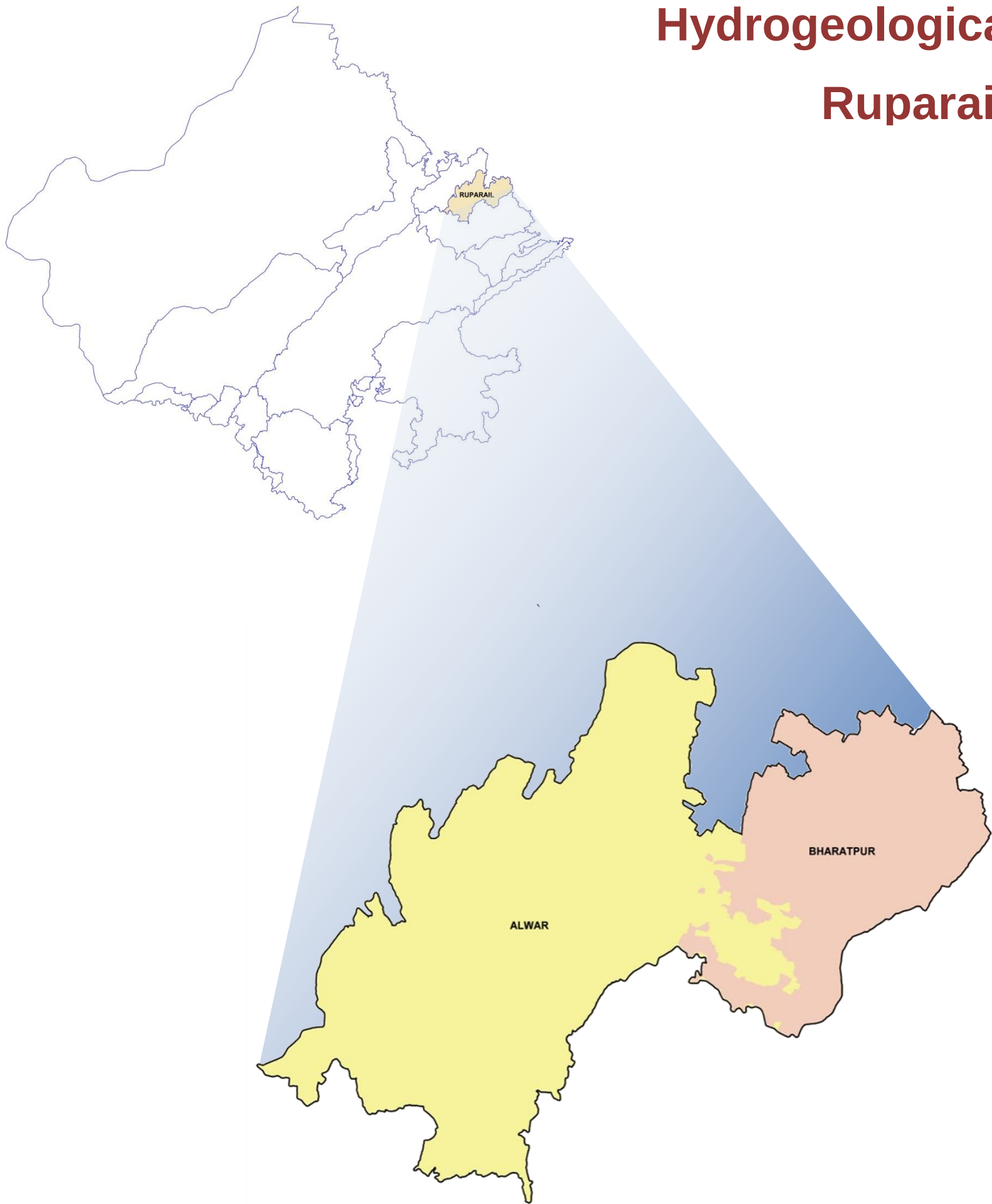
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Ruparail River Basin



Contents:

List of Plates	Title	Page No.
Plate I	Administrative Map	2
Plate II	Topography	4
Plate III	Rainfall Distribution	4
Plate IV	Geological Map	6
Plate V	Geomorphological Map	6
Plate VI	Aquifer Map	8
Plate VII	Location of Ground Water Monitoring Stations	8
Plate VIII	Location of Exploratory Wells	10
Plate IX	Depth to Water Level (Pre-Monsoon 2010)	10
Plate X	Water Table Elevation (Pre-Monsoon 2010)	12
Plate XI	Water Level Fluctuation (Pre-Post Monsoon 2010)	12
Plate XII	Electrical Conductivity Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIII	Chloride Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIV	Fluoride Distribution (Average Pre-Monsoon 2005-09)	16
Plate XV	Nitrate Distribution (Average Pre-Monsoon 2005-09)	16
Plate XVI	Depth to Bedrock	18
Plate XVII	Map of Unconfined Aquifer	18
Plate XVIII	Index Map Showing Alignment of Cross Sections	20
Plate XIX	Cross Section Along A-A'	22
Plate XX	Cross Section Along B-B'	22
Plate XXI	Cross Section Along C-C'	24
Plate XXII	Cross Section Along D-D'	24
Plate XXIII	3D Model	26
Glossary of terms		27

ADMINISTRATIVE SETUP

RUPARAIL RIVER BASIN

Location:

The Ruparail River Basin is located in the northeastern part of Rajasthan. It stretches between 27° 09' 22.24" to 27° 55' 10.35" North latitude and 76° 14' 42.36" to 77° 21' 34.47" East longitudes. The basin extends in a broadly W-E direction and is bounded by the Sabi River Basins in the northwest side and the Banganga River Basin in the southeast. The northern border is shared with Haryana State. The basin extends over parts of Bharatpur and Alwar Districts. The total catchment area of the Basin is very small and extends over an approximate area of 3,809.6 km².

River Ruparail, also known as Barah, rises in the Udainath hills of Thanagazi Reserved Forest (RF), Alwar District. It traverses these hills northwards, turning towards the east and northeast before disappearing in Bharatpur District. It flows first through hills and subsequently through plains nearly up to Kusalpur in Bharatpur district. It has a total length of about 104 km. A number of smaller streams rise from the various sub-ranges of the above-mentioned RF hills, e.g. the Narainpur, Golari, Sukri, Shanganga and Nalakroti Rivers and contribute to Ruparail flow.

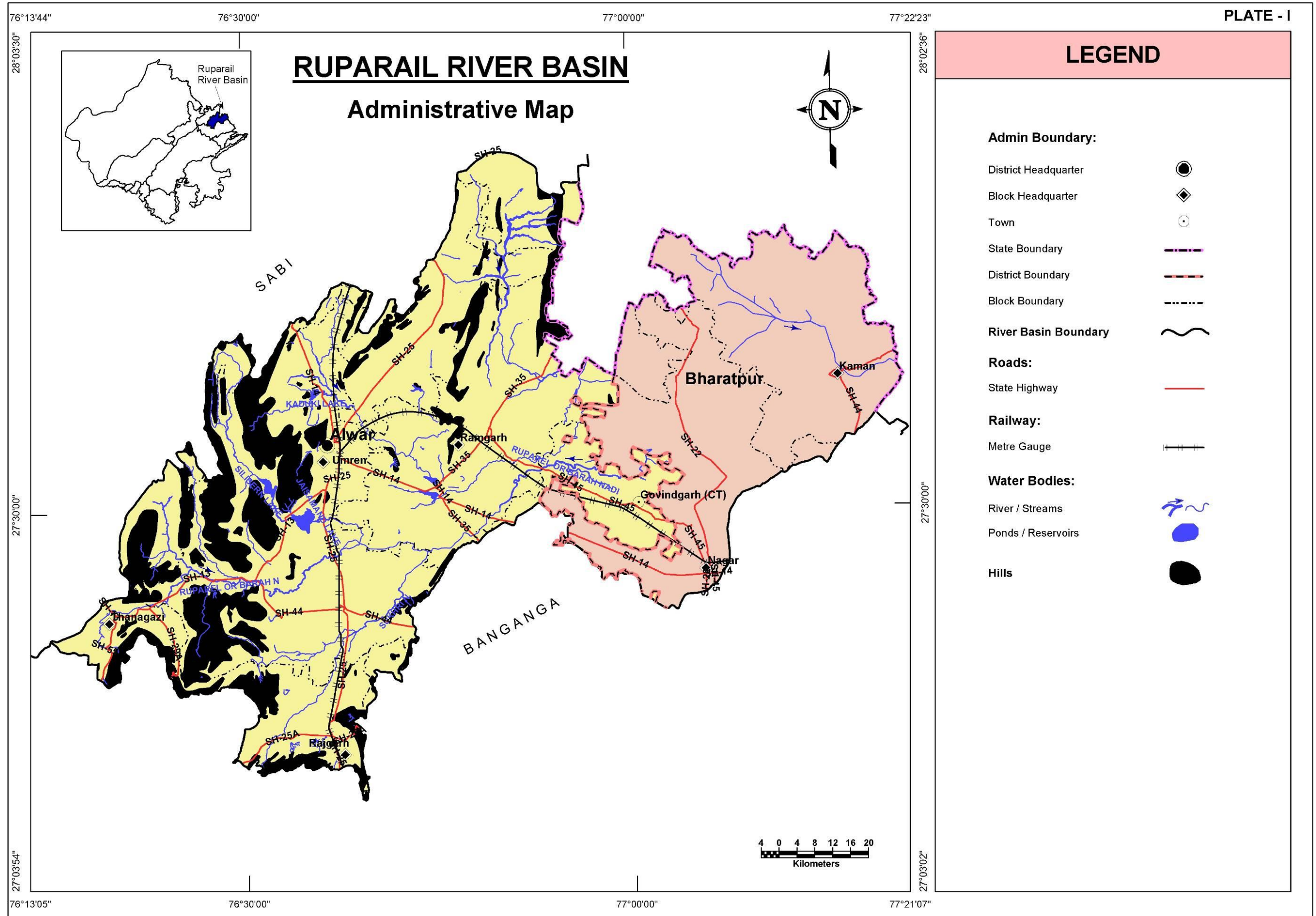
Administrative Set-up:

Administratively, Ruparail River Basin extends over parts of Bharatpur and Alwar districts encompassing 15 Blocks divided into 984 towns and villages. Major part of the river flows through Alwar district where its catchment is about 71% of basin area whereas the rest falls within the Bharatpur district.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Alwar	2,694.5	70.7	12	582
2	Bharatpur	1,115.1	29.3	3	402
Total		3,809.6	100.0	15	984

Climate:

The basin is located on the northeastern part of Rajasthan sharing its boundary with Haryana. It falls within the semi-arid to sub-humid regions but witnesses extreme temperatures during the hottest months which are May and June. Coldest months are December and January where temperatures sometimes reach near-zero levels. Incidentally, India's highest ever recorded temperature of 50.6 °C was measured at Alwar on 10 May 1956. The rain are mostly received during monsoon season i.e., from July through September, and mean annual rainfall over the basin was computed as 576mm.



TOPOGRAPHY

RUPARAIL RIVER BASIN

The western part is hilly where the river originates, and then the terrain slopes towards east and then to northeast. The hills mostly belong to the Aravali ridges are locally curvilinear representing dome and basin structures but follow a broad southwest – northeast trend. The river has an inland drainage system and ends up in the depression around Bharatpur city. The elevation ranges from a minimum of 179m in Bharatpur district to a maximum of about 716m in Alwar district. Ghana bird sanctuary area is actually a regional depression with no exit that leads to retention of surface water flows.

Table: District wise minimum and maximum elevation

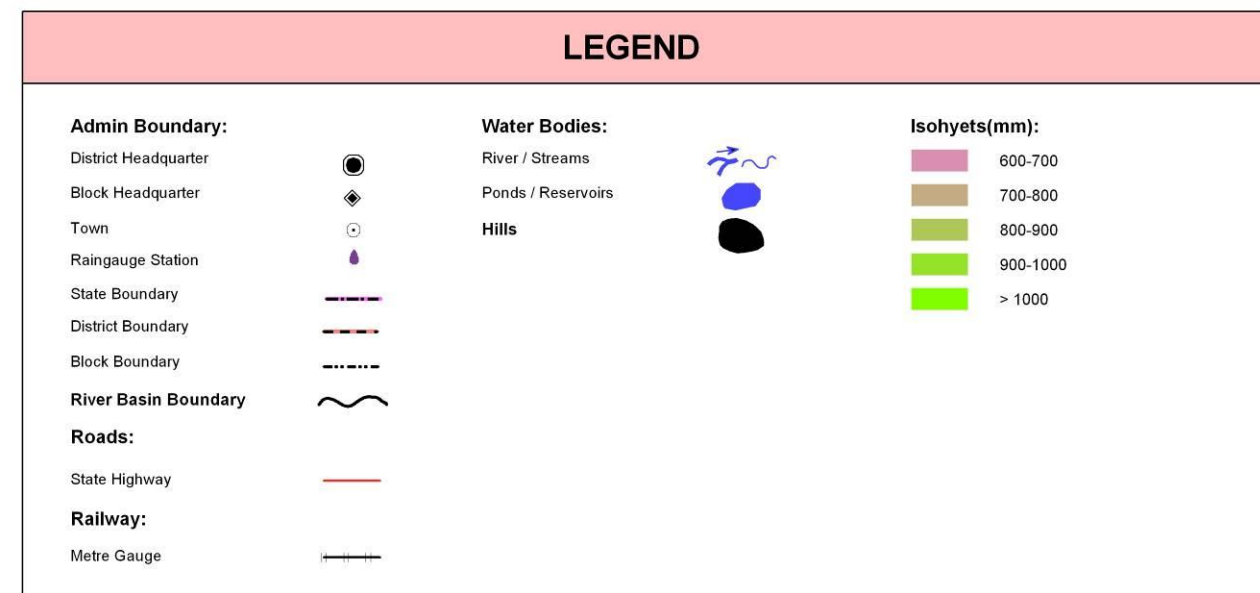
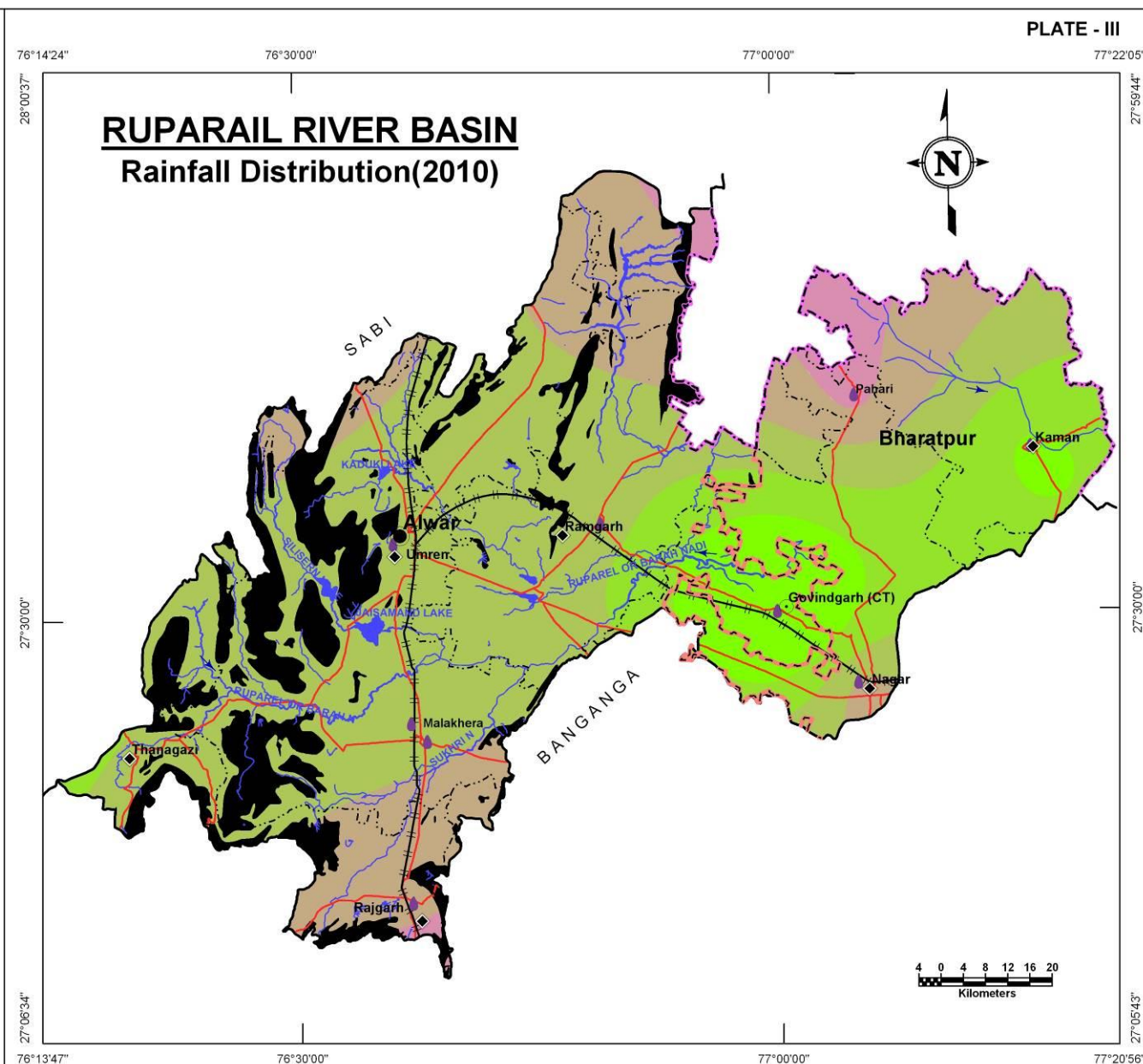
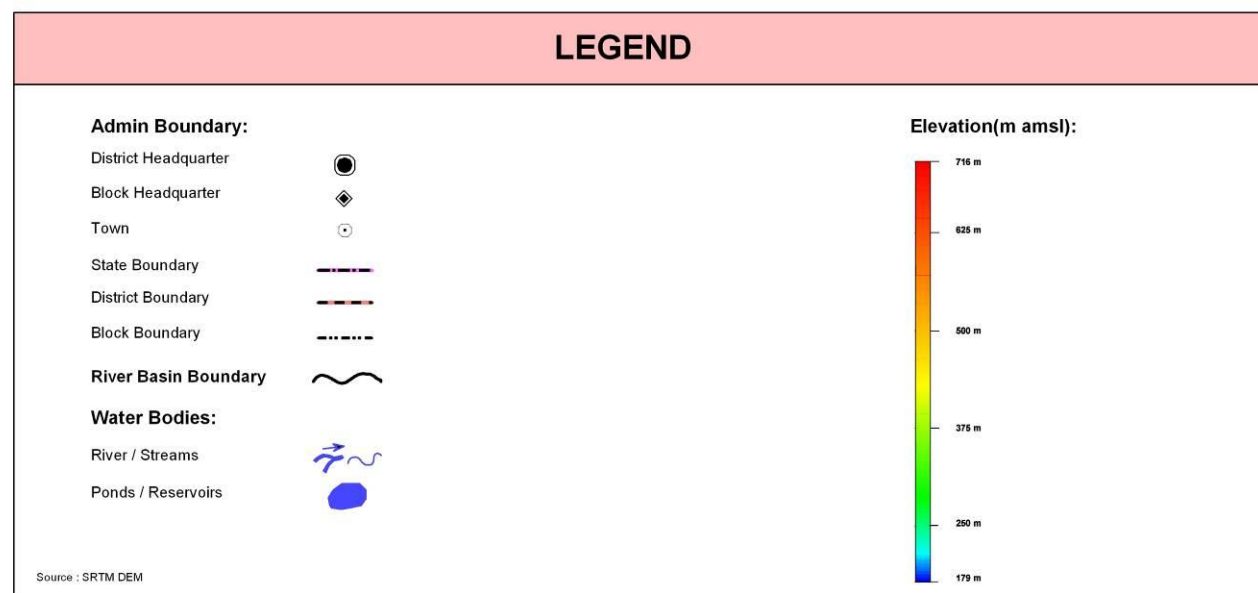
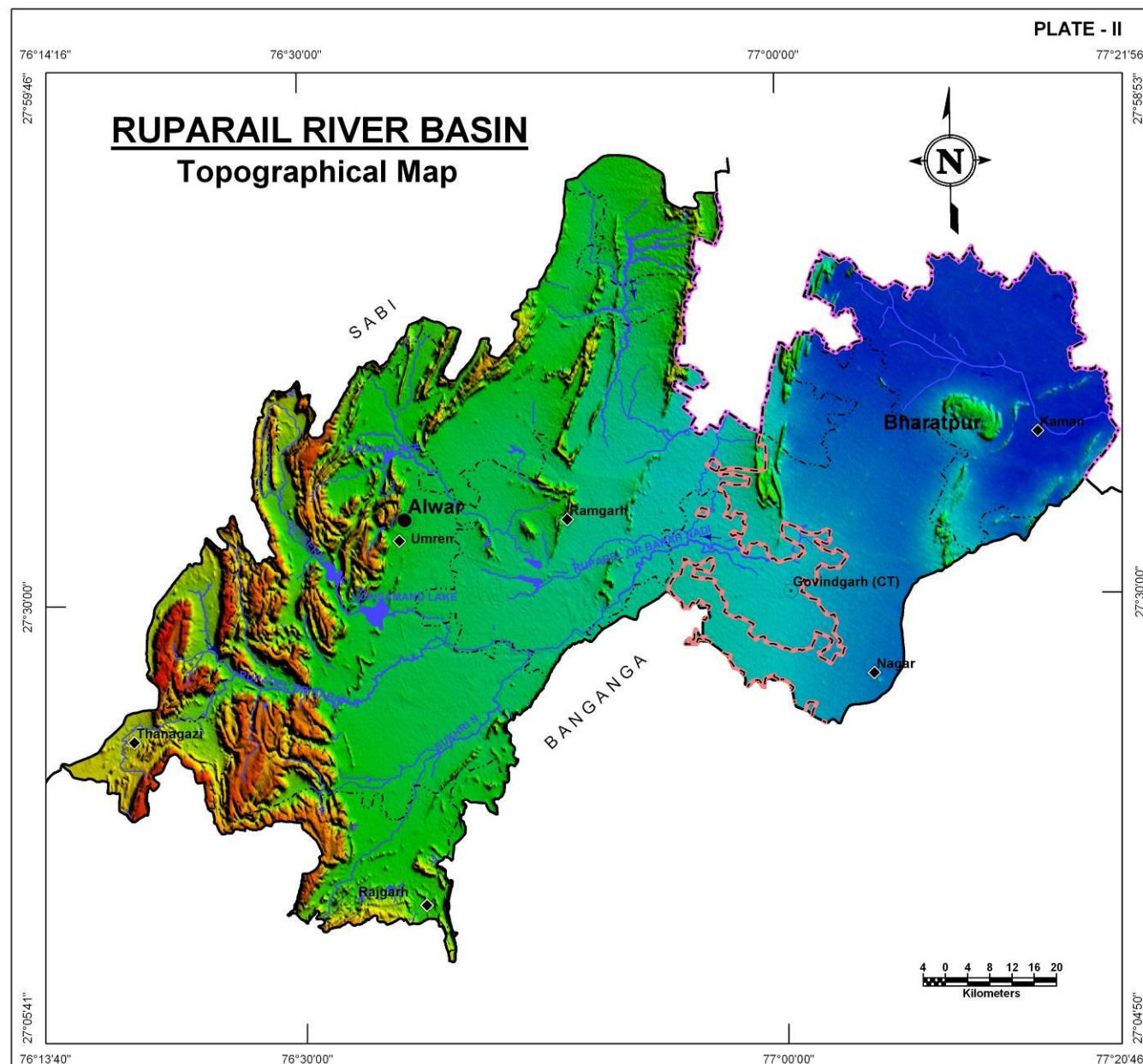
S. No.	District Name	Min Elevation (m amsl)	Max. Elevation (m amsl)
1	Alwar	204.0	716.4
2	Bharatpur	179.4	366.2

RAINFALL

The general distribution of rainfall across the Ruparail River Basin can be visualized from isohyets presented in the Plate – III where the basin seems to have received good rainfall in the year 2010. The total annual rainfall was in excess of 1200mm in basin as recorded by the Govindgarh met station followed by Kaman. The lowest total annual rainfall in the basin was recorded at the Pahari station which accounted for 669mm. The average annual rainfall computed based on available station data is about 871 mm. The rainfall data for available rain gauge stations is presented below.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Alwar	781.0	76.0	857.0
2	Govindgarh	1,239.0	25.0	1,264.0
3	Kaman	965.0	57.0	1,022.0
4	Malakhera	767.0	59.0	826.0
5	Nagar	687.0	57.0	744.0
6	Pahari	603.0	66.0	669.0
7	Rajgarh	634.0	80.0	714.0
8	Ramgarh	806.7	67.0	873.7



GEOLOGY

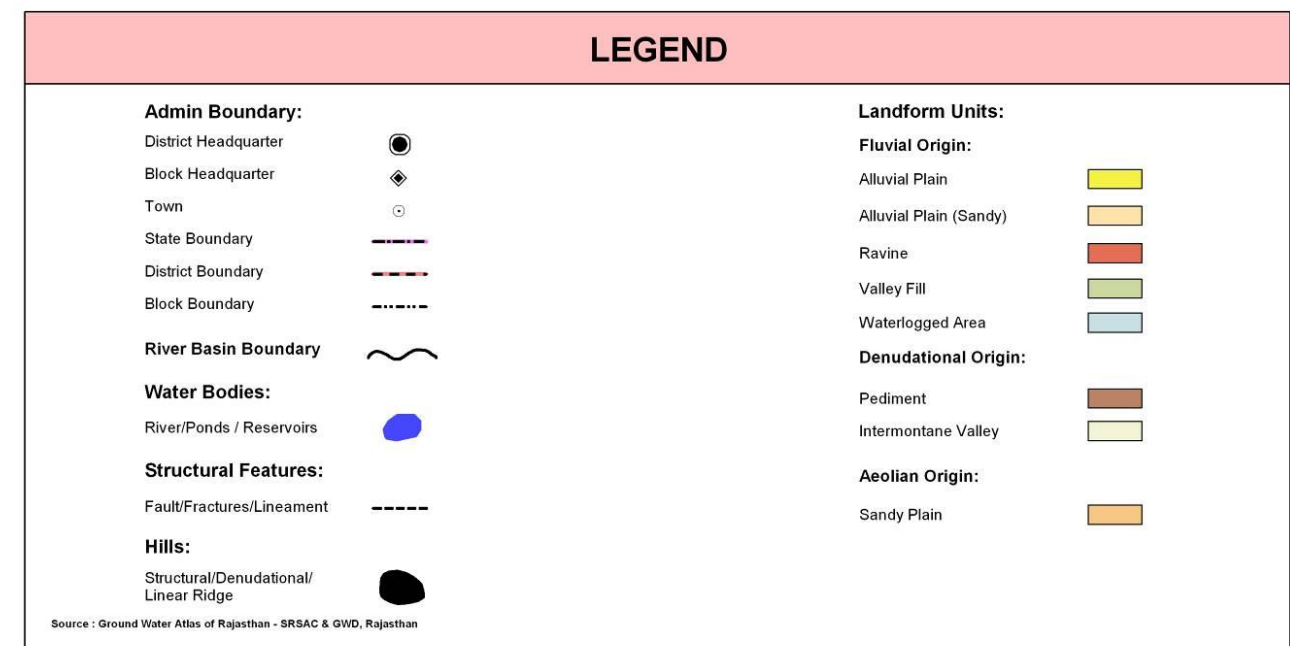
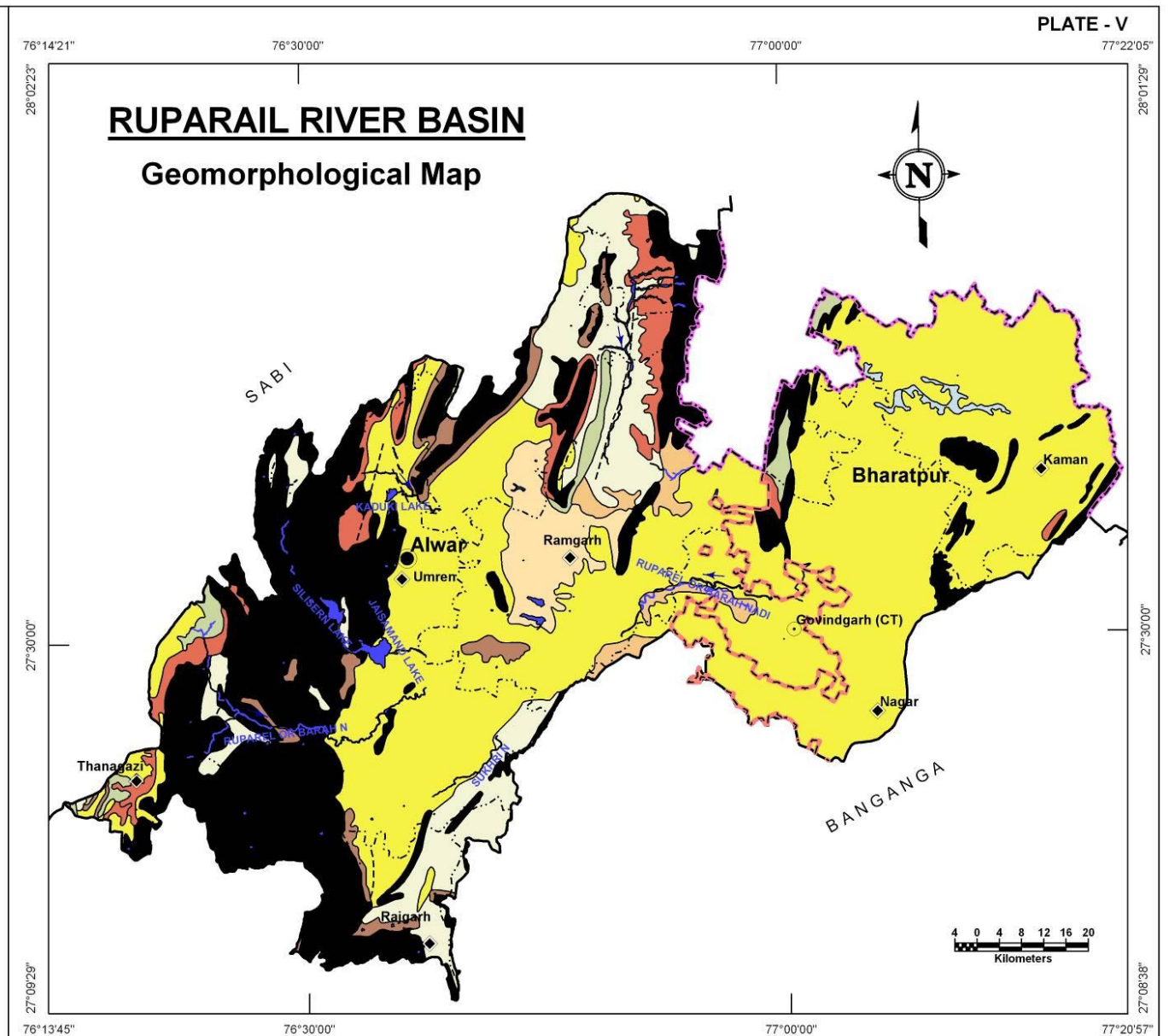
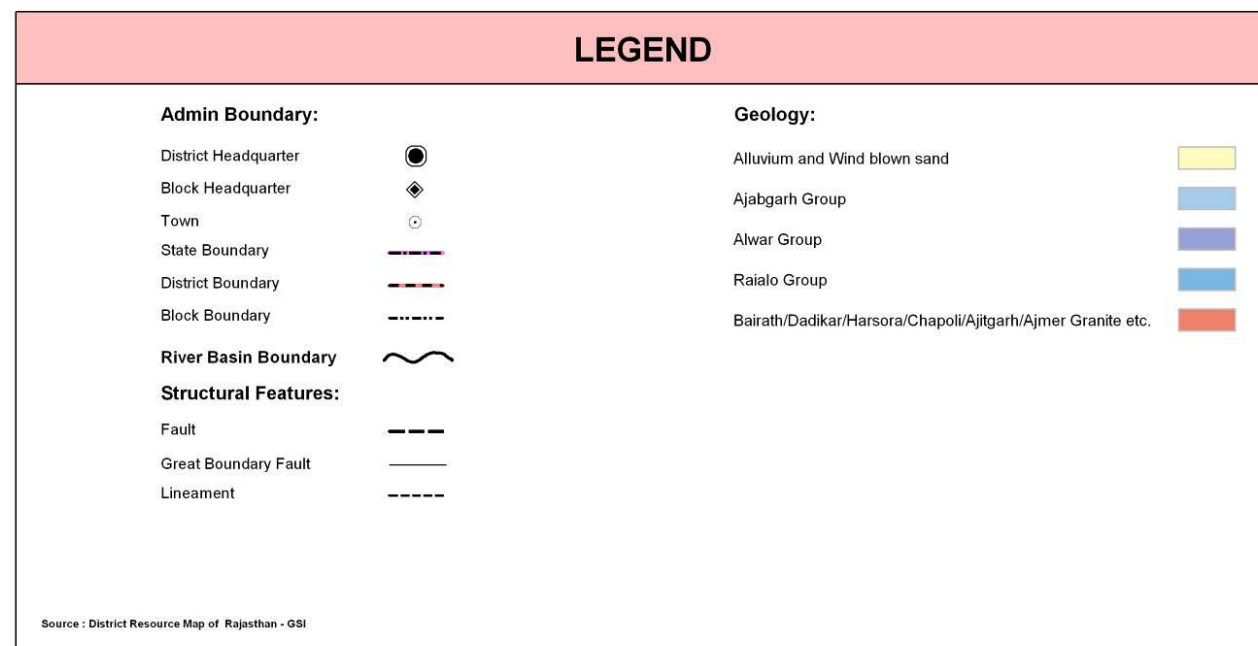
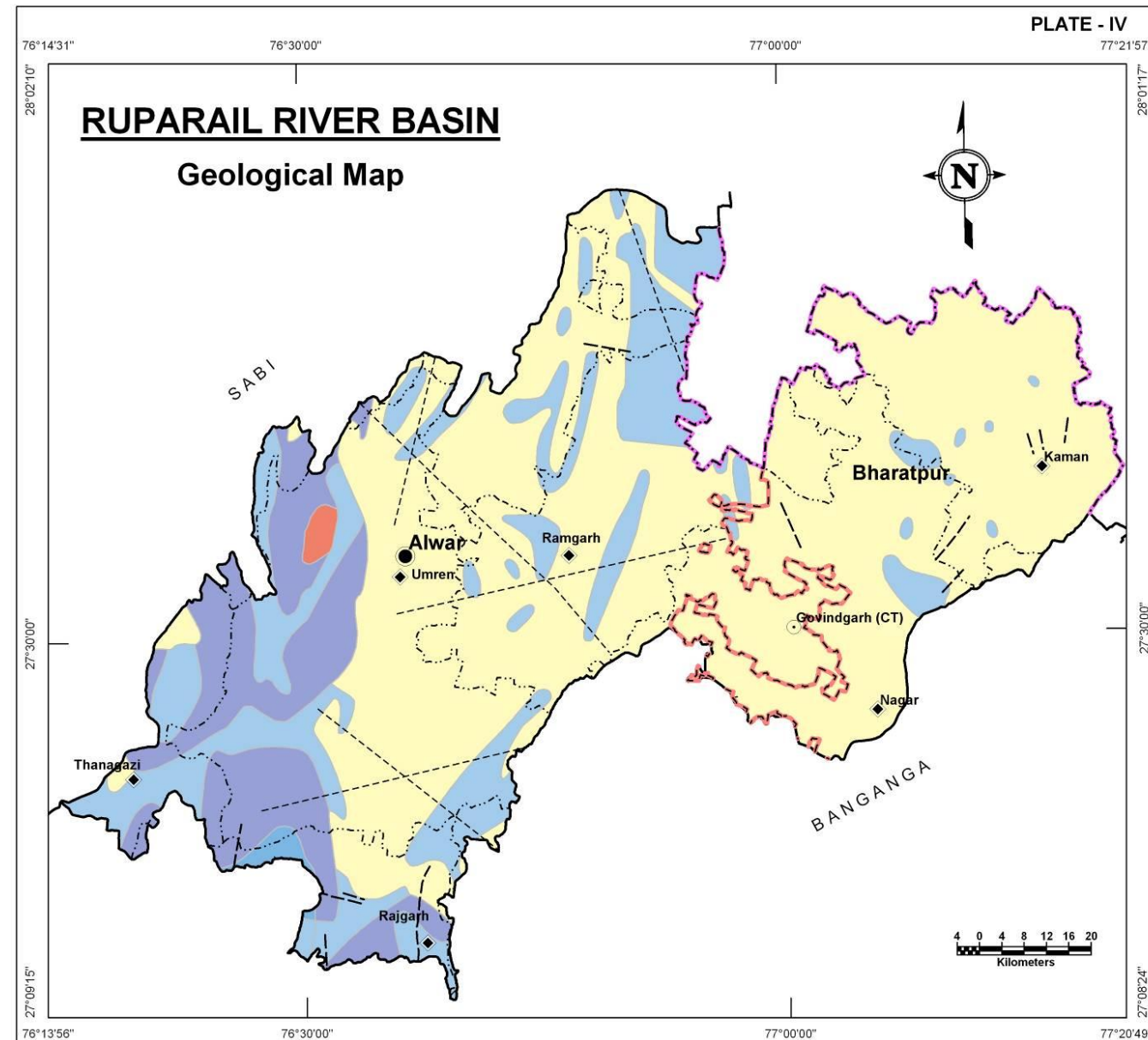
RUPARAIL RIVER BASIN

The area covered by the basin is very small (about 3800 sq km) so geologic variation in the basin is also very limited. The alluvium (sand, clay, kankar and coarse colluvial material) and wind-blown sand occupies significant area in the central and eastern part of the basin whereas Delhi Super Group rocks occupy the rest of the area in the west and also appearing sporadically in the alluvial terrain along with Raialo Group of rocks represented by dolomites, quartzites etc.

Age	Super-Group	Group/ Formation	Rock Types
Sub-Recent To Recent	Alluvium	Alluvium	Alluvial sand, Clay, Kankar with subordinate colluvial gravel, Pebbles, cobbles etc., and wind-blown sand or Dune sand
-----X-----X-----Unconformity-----X-----X-----X-----			
Upper Precambrian To lower Cambrian	Delhi	Ajabgarh	Schists, phyllites, Quartzite and dolomites
		Alwar	Quartzite, Conglomerates, grits With subordinate shale and calcareous bands, Meta dolerites, epidiorites
		Raialo	Dolomites, Limestone Sandstones/ quartzite

GEOMORPHOLOGY

Origin	Landform Unit	Description
Aeolian	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
Denudational	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
	Water logged/ Wetland	Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

RUPARAIL RIVER BASIN

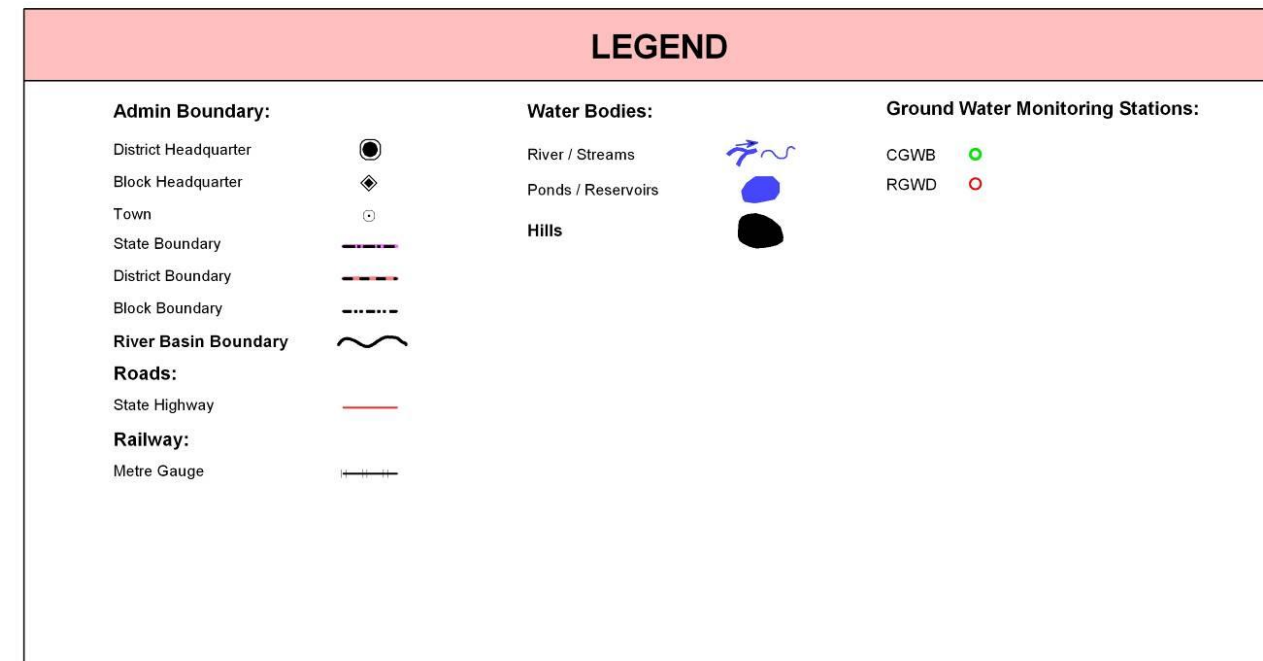
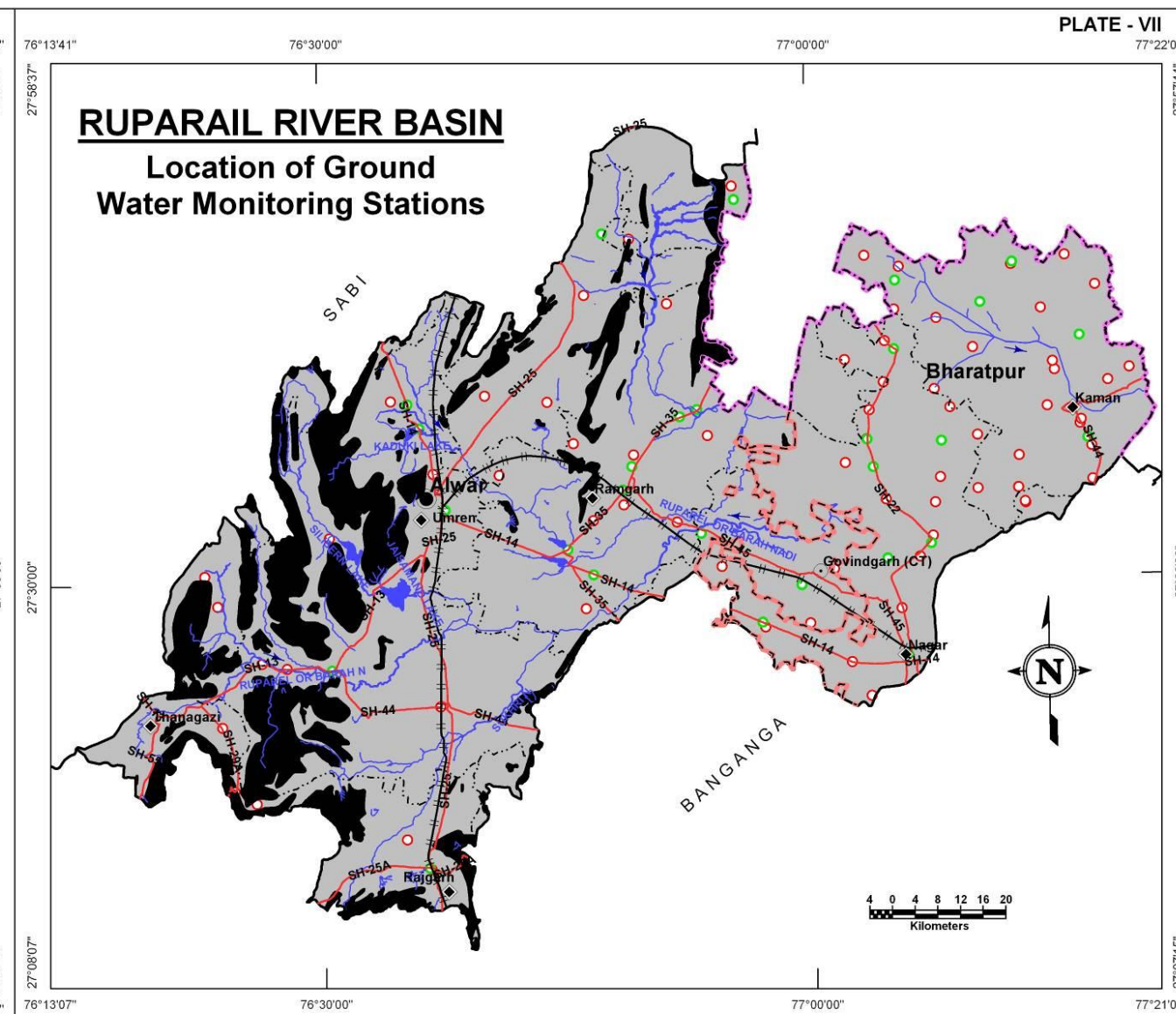
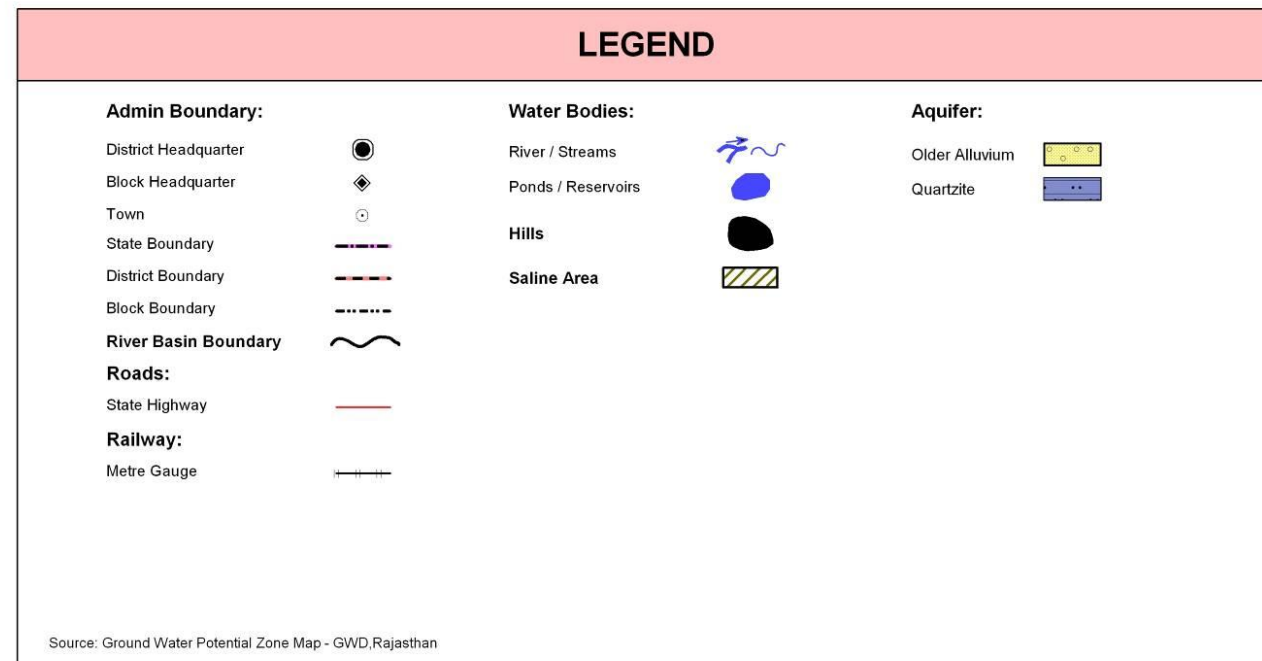
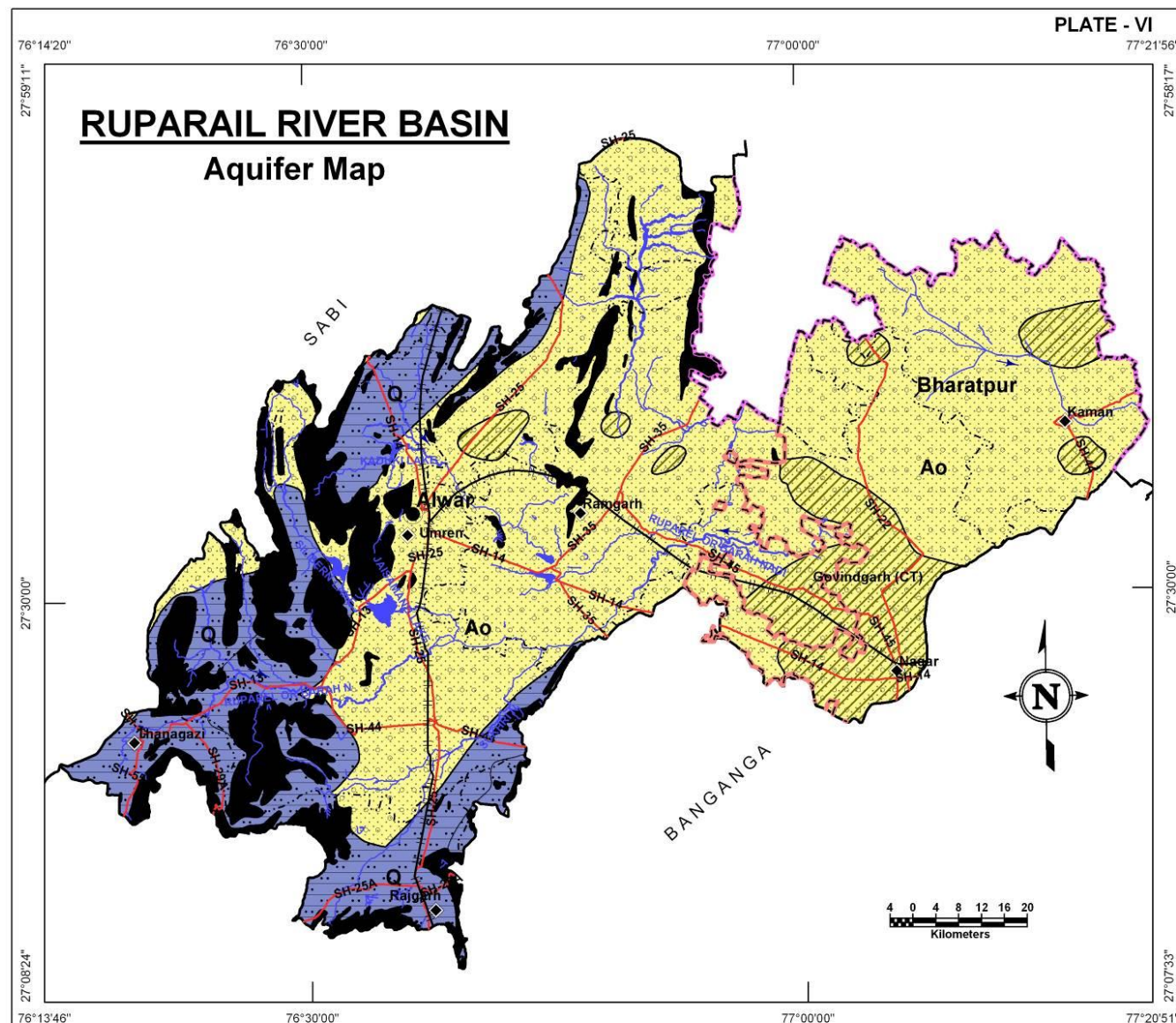
Weathered and fractured quartzites and alluvium constitute principal aquifers in the basin. Older alluvium occupies about 70% of the basin area forming it the most important aquifer, however, the quality of water in some parts of the aquifer is not of potable quality. The quartzite aquifers are seen in hilly region and its vicinity which generally contains fresh water occupies about 16% of the basin area.

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Older Alluvium	2,696.1	70.7	This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.
Quartzite	615.7	16.2	Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.
Non Potential Zone	497.8	13.1	Hills and reserve forests.
Total	3,809.6	100.0	

LOCATION OF GROUND WATER MONITORING WELLS

The basin has a well distributed network of large number of ground water monitoring stations (104) in the basin owned by RGWD (76) and CGWB (28); and an additional 116 wells have been recommended to be added to network to effectively monitor ground water quality (97 additional wells) and water level (19 additional wells) in the basin.

District Name	CGWB	RGWD	Total	Recommended additional wells for optimization of monitoring network	
				Water Level	Water Quality
Alwar	15	31	46	19	63
Bharatpur	13	45	58	-	34
Total	28	76	104	19	97



LOCATION OF EXPLORATORY WELLS

In all there are 162 exploratory wells present in the basin drilled in the past by RGWD (152) and CGWB (10) that form the basis for delineation of sub-surface aquifer system. A perusal of Plate – VII reveals that the density of wells is more in alluvial areas than in hilly hardrock areas. Large number of exploratory wells are seen around Alwar city.

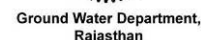
District Name	CGWB	RGWD	Total
Alwar	4	117	121
Bharatpur	6	35	41
Total	10	152	162

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin is less than 30 meters below ground level, as seen in most of the eastern and western parts of the basin. There is a significant deepening of water levels in the central part around Alwar city and further southwards as indicated by >30m bgl water level contours reaching depths of more than 40m bgl. The shallow depth to ground water levels is seen in the eastern most part around Kaman and Bharatpur where the water occurs at depths of less than 10m bgl.

Depth to water level (m bgl) Pre Monsoon - 2010	District wise area coverage (sq km)*		Total Area (sq km)
	Alwar	Bharatpur	
<10	12.5	441.2	453.7
10-20	738.7	624.1	1,362.8
20-30	601.8	49.9	651.7
30-40	536.2	-	536.2
>40	307.4	-	307.4
Total	2,196.6	1,115.2	3,311.8

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON 2010)

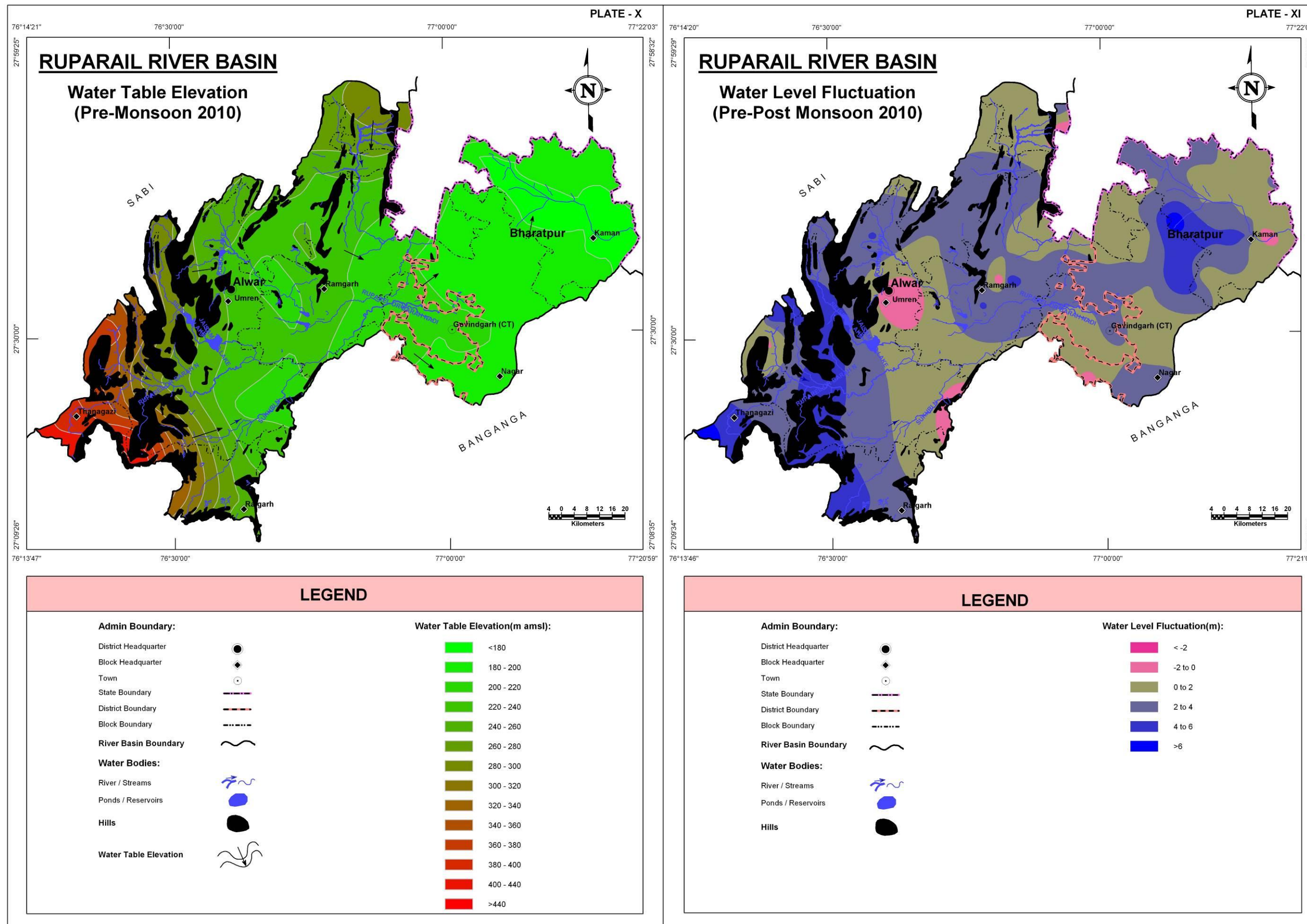
The flow directions marked on water table elevation contour map (Plate – X) indicate an initial southwest to northeast flow and another regional sub-surface flow from north to south. These two flow directions meet in the centre of the basin and ultimately follow an eastwards flow towards Bharatpur. The water table elevations show very steep gradients in the western hilly part which more gradual and sluggish towards east. The variation between minimum and maximum water table elevations is quite large, being about 180m amsl to about 440m amsl resulting into a net head difference of 260m. For major part of the basin, the water table elevation varies in the general range of 180m amsl to 240m amsl.

Water Table Elevation (m amsl) Pre Monsoon - 2010	District wise area coverage (sq km)		Total Area (sq km)
	Alwar	Bharatpur	
< 180	-	223.6	223.6
180 - 200	47.8	809.3	857.1
200 - 220	465.0	82.3	547.3
220 - 240	690.4	-	690.4
240 - 260	331.4	-	331.4
260 - 280	204.4	-	204.4
280 - 300	186.3	-	186.3
300 - 320	55.0	-	55.0
320 - 340	55.4	-	55.4
340 - 360	50.8	-	50.8
360 - 380	51.8	-	51.8
380 - 400	33.3	-	33.3
400 - 440	24.6	-	24.6
> 440	0.4	-	0.4
Total	2,196.6	1,115.2	3,311.8

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

Water level fluctuation map is presented in Plate – XI. Perusal of the map reveals that the hilly areas in the western part of the basin show greater rise in the water table in post-monsoon season as compared to pre-monsoon water levels, whereas the flatter alluvial areas show lesser fluctuation. The hilly areas with hardrock aquifers had shown a general rise of 2-6m and in the western extremity reaching a high of 6m. There is an unusually high of 6m west of Kaman. The alluvial areas have shown a general fluctuation range of 0-4m and on some occasions showed a fall in water level (in post monsoon period) by more than 2m as seen around Kaman and especially around Alwar, possibly because of ground water extraction exceeding monsoon recharge.

District Name	District wise area coverage (sq km) within fluctuation range (m)						Total Area (sq km)
	< -2	-2 to 0	0 to 2	2 to 4	4 to 6	> 6	
Alwar	0.4	88.5	824.4	1,061.4	213.2	8.6	2,196.5
Bharatpur	-	12.1	539.3	441.9	113.1	8.8	1,115.2
Total	0.4	100.6	1,363.7	1,503.3	326.3	17.4	3,311.7



ELECTRICAL CONDUCTIVITY DISTRIBUTION

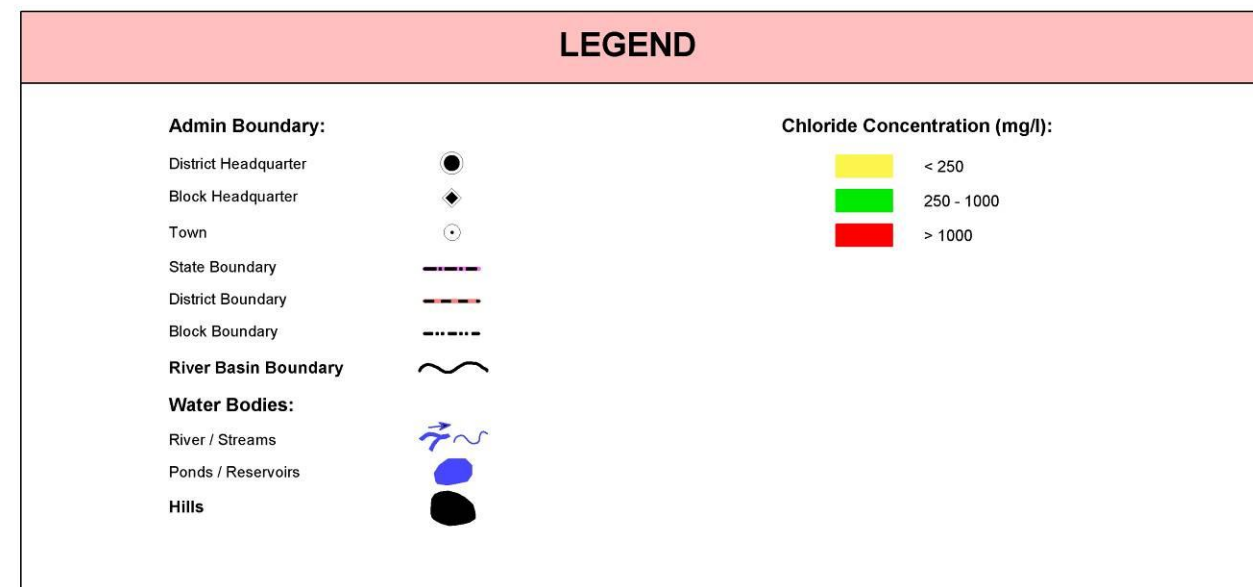
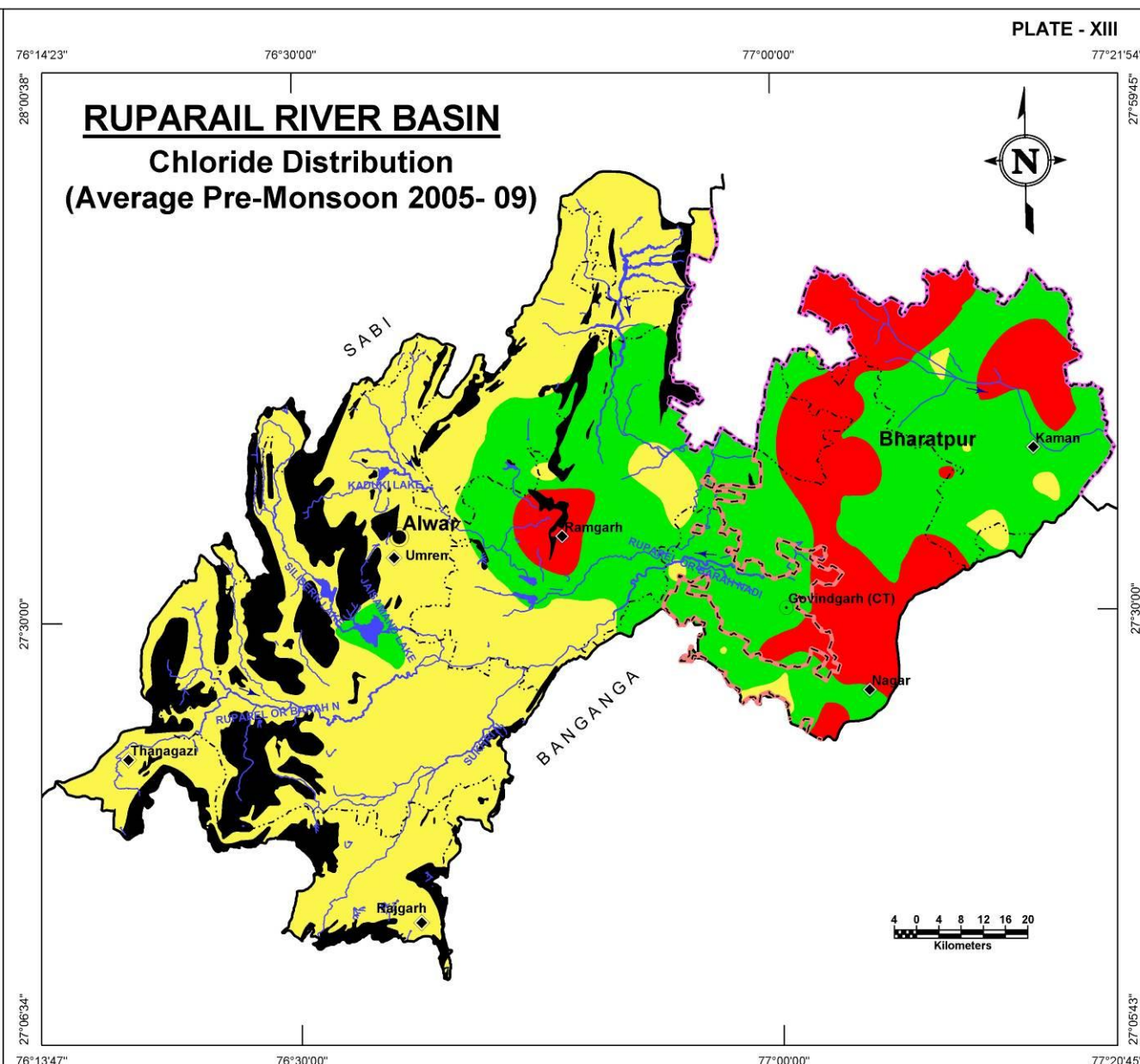
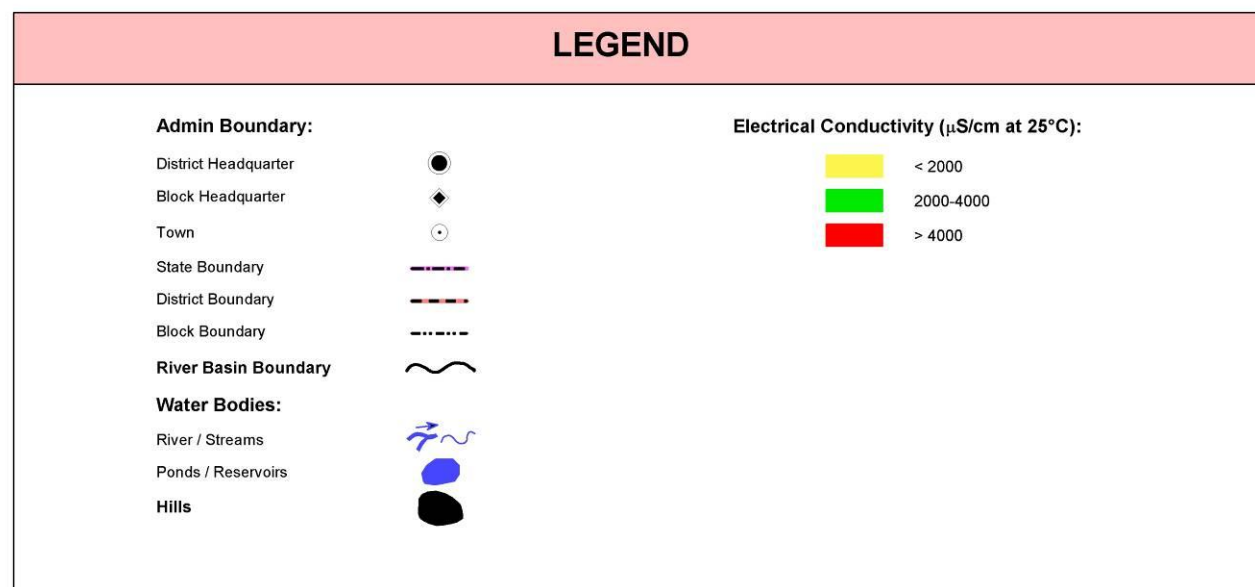
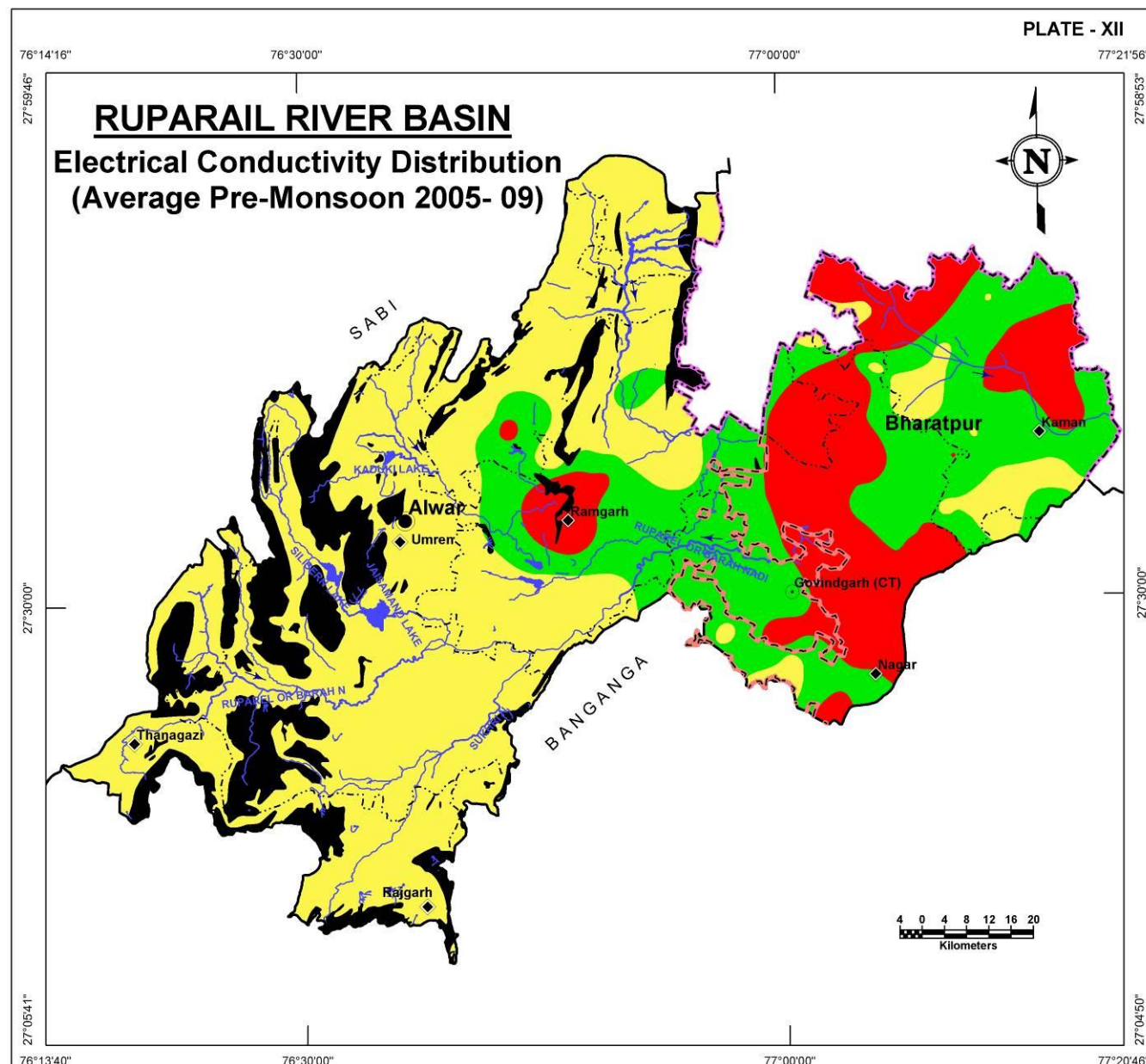
The Electrical Conductivity (at 25°C) distribution map is presented in Plate – XII reveals an in general low values (EC <2000 $\mu\text{S}/\text{cm}$) in the western part of the basin which makes it suitable for domestic purposes. From the central part of the basin (east of Alwar city) till the edge of the basin in the east the EC values gradually show a rise and large patches of high salinity could be seen in Bharatpur district. Interestingly, the low EC areas largely correspond to hardrock aquifer areas and the high salinity (EC >4000 $\mu\text{S}/\text{cm}$) is seen in alluvial aquifers especially the areas where the Ruparail river disappears in the topographic low around Bharatpur. The low, moderate and high EC areas occupy 55.4%, 26.7% and 17.9 % of the basin areas respectively.

Electrical Conductivity Ranges (μS/cm at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Alwar		Bharatpur		
	Area	% age	Area	% age	
< 2000	1,724.7	78.5	109.1	9.8	1,833.8
2000-4000	374.6	17.1	510.1	45.7	884.7
> 4000	97.2	4.4	496.0	44.5	593.2
Total	2,196.5	100.0	1,115.2	100.0	3,311.7

CHLORIDE DISTRIBUTION

High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. The chloride concentration map (Plate – XIII) exhibits a similar distribution pattern as to that of Electrical Conductivity. A predominantly yellow colour (<250 mg/l) in the map suggests that from chloride perspective, the ground water is suitable for all purposes and together with moderately high concentration (250 – 1000 mg/l), these areas occupy about 84% of basin area and the rest of the area (about 16%) has high chloride concentration (>1000 mg/l) in ground water. The hilly aquifers areas formed in quartzites broadly correspond to low chloride concentration areas whereas the alluvial aquifers seem to have, in general, higher chloride concentration in ground water.

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Alwar		Bharatpur		
	Area	% age	Area	% age	
< 250	1,543.6	70.2	28.2	2.5	1,571.8
250 – 1000	572.7	26.1	634.0	56.9	1,206.7
> 1000	80.2	3.7	453.0	40.6	533.2
Total	2,196.5	100.0	1,115.2	100.0	3,311.7



FLUORIDE DISTRIBUTION

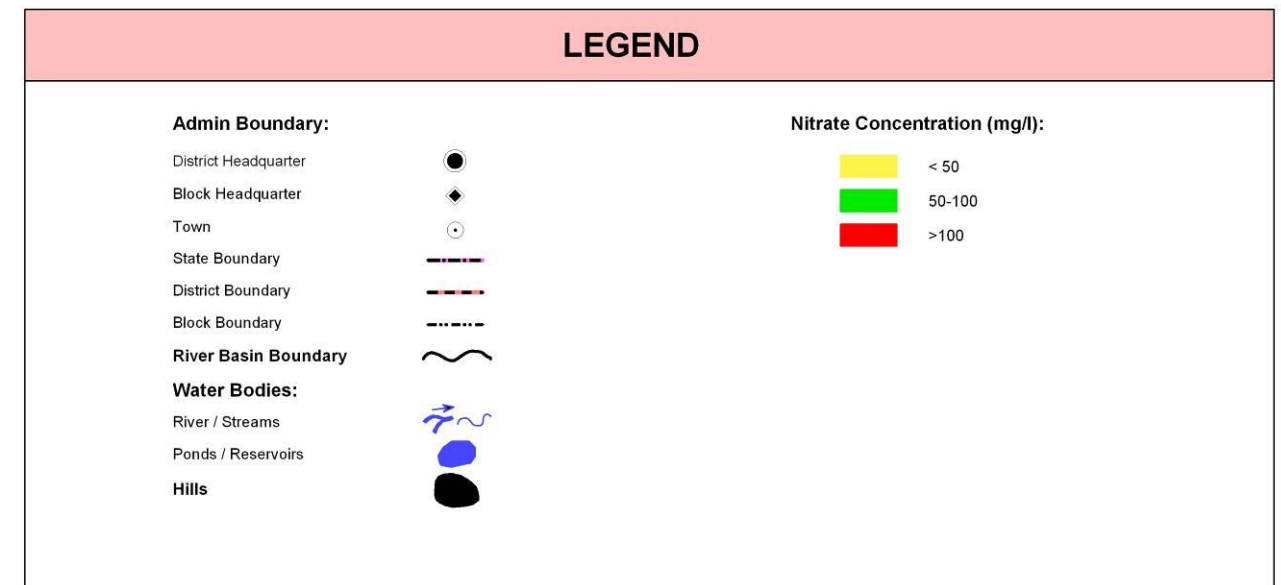
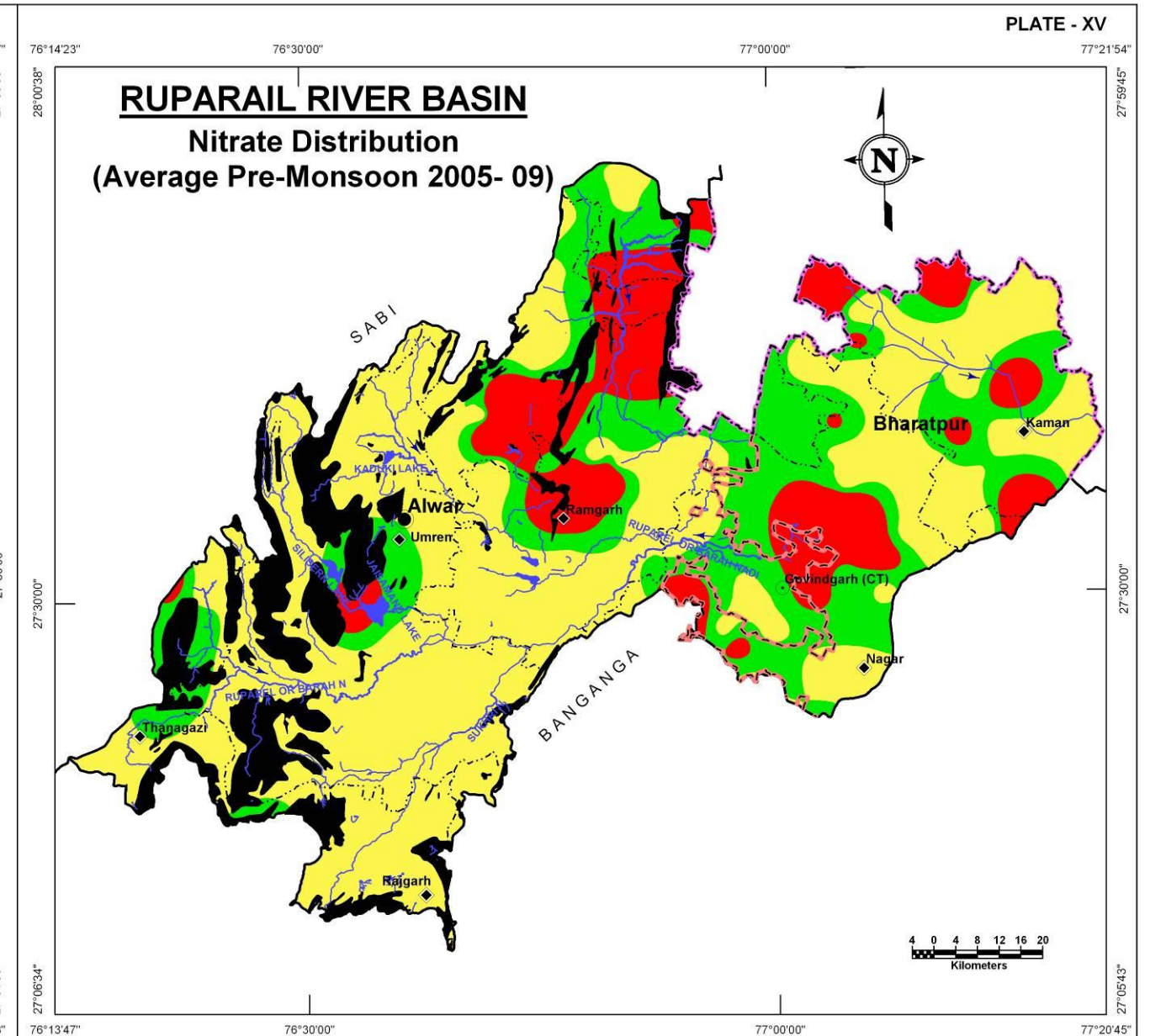
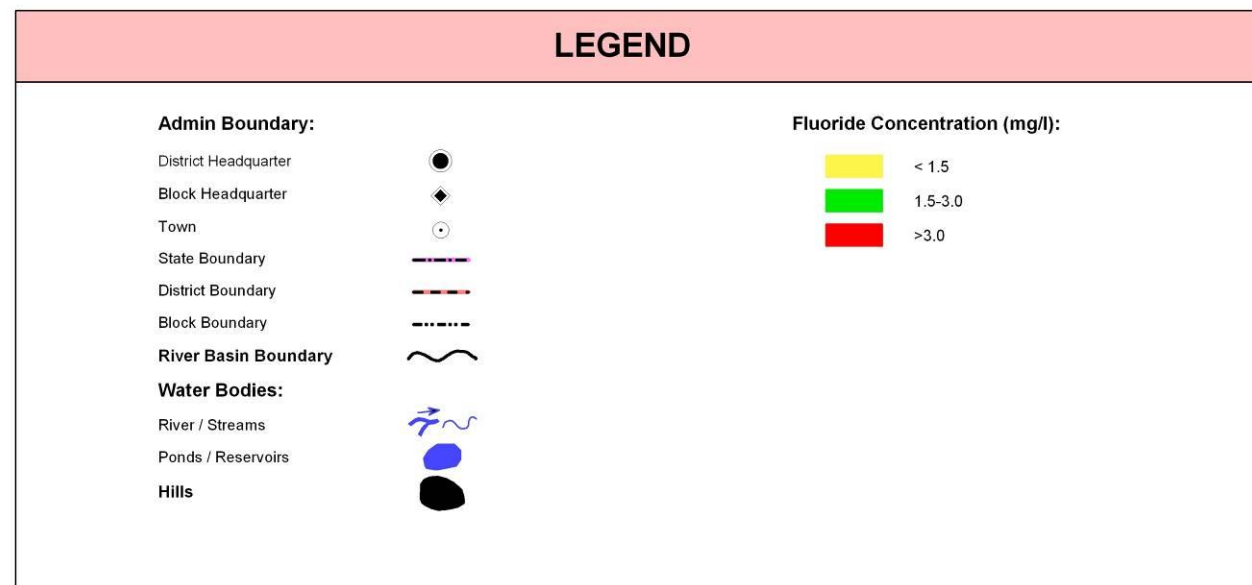
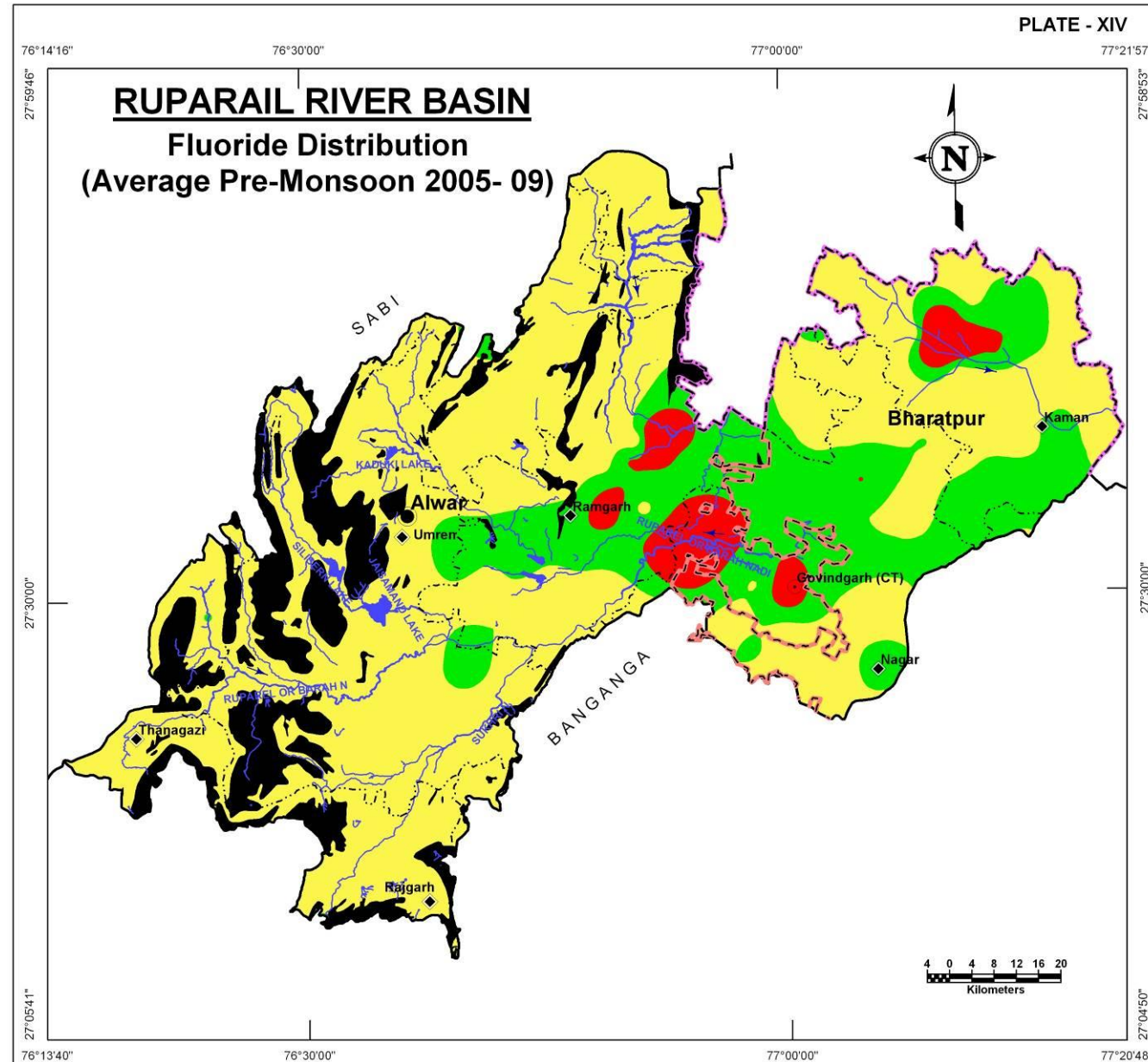
The Fluoride concentration map for Ruparail River Basin (Plate XIV) shows some scattered patches where very high Fluoride concentration (>3 mg/l) was recorded and these occupy an area of approximately 5.4% area of Alwar district in its easternmost part adjoining Bharatpur, whereas the high fluoride patch in Bharatpur district is present in its northern part. Overall, the high fluoride areas occupy about 5% of the basin area. Moderate Fluoride concentration (1.5 – 3.0 mg/l) areas surround the high fluoride areas and together in both districts occupy about 25% area. The rest of the area (approximately 70%) has shown very low concentration of fluoride in ground water samples which makes it suitable for all purposes.

Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Alwar		Bharatpur		
	Area	% age	Area	% age	
<1.5	1,726.5	78.6	603.5	54.1	2,330.0
1.5-3.0	350.5	16.0	462.4	41.5	812.9
>3.0	119.5	5.4	49.3	4.4	168.8
Total	2,196.5	100.0	1,115.2	100.0	3,311.7

NITRATE DISTRIBUTION

Plate XV shows distribution of Nitrate in ground water which is important to assess the ground water's suitability for agriculture. The distribution pattern does not show any direct relation to aquifers since several isolated high Nitrate concentration (>100 mg/l) patches are seen distributed over the area in the northern hilly areas of Alwar district as well as central and eastern parts within alluvial areas of the basin. Overall, 16.6% of the basin area is unsuitable for agriculture, however some part of this high Nitrate concentration areas fall within hilly (non-agriculture) areas also. This leaves about 83.4% of the basin area falling within low to moderate (i.e., <100 mg/l).

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)				Total Area (sq km)
	Alwar		Bharatpur		
	Area	% age	Area	% age	
< 50	1,427.6	65.0	465.3	41.7	1,892.9
50-100	451.7	20.6	416.5	37.4	868.2
> 100	317.2	14.4	233.4	20.9	550.6
Total	2,196.5	100.0	1,115.2	100.0	3,311.7



DEPTH TO BEDROCK

The entire area of the Ruparail River basin is underlain by the hard rocks at different depths. The major rocks types occurring are Slate and Quartzite. These rocks are overlain by alluvial deposits of sand, clay, silt, kankar with subordinate colluvial, gravels, pebbles, cobbles and windblown sand and dune sand. The depth to bed rock defines the sub surface topography of the occurrence of massive hard rock beneath alluvial deposits and weathered and fractured hard rocks that are significant from aquifer mapping perspective.

On perusal of the map (Plate – XVI) of depth to bedrock (in meters above ground level) it is apparent that the bedrock is at deeper levels in eastern part of the river basin which gradually occurs at shallower depths in the western part and finally exposed as hills, further west and north. In the western part the hills apparently have steeply dipping slopes because the bedrock encountered in the adjacent areas also is deep, ranging between 60 to 80m below ground level. In the eastern part of Alwar district and in most part of Bharatpur district, the depth to bedrock gradually increases towards east attaining depths of more than 140m bgl.

Depth to Bedrock (m bgl)	District wise area coverage (sq km)		Total Area (sq km)
	Alwar	Bharatpur	
< 60	370.0	-	370.0
60-80	1,310.2	-	1,310.2
80-100	387.2	250.6	637.8
100-120	127.0	632.1	759.1
120-140	2.1	206.0	208.1
> 140	-	26.5	26.5
Total	2,196.5	1,115.2	3,311.7

UNCONFINED AQUIFER

Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas.

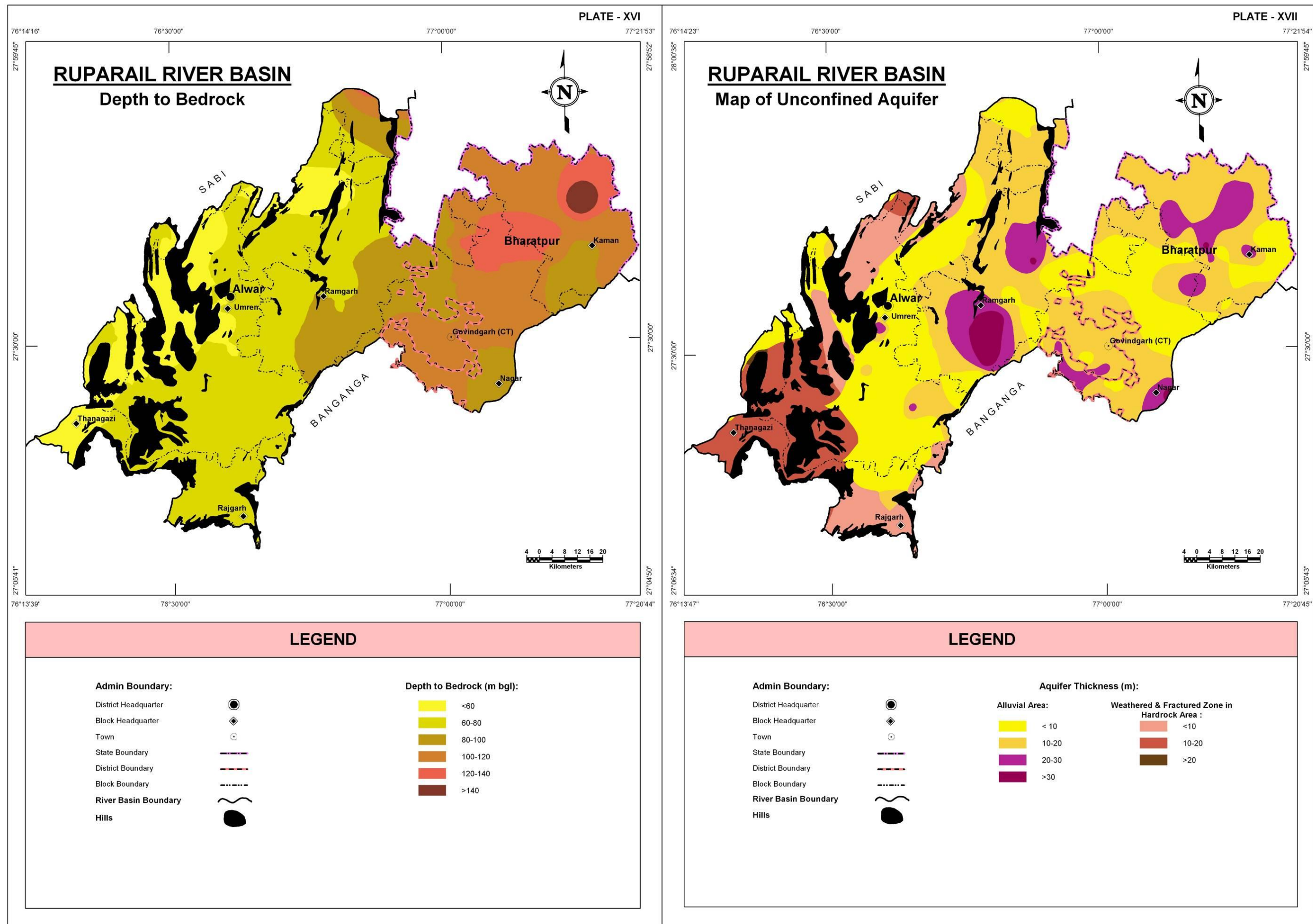
Except for hardrock exposure areas, this aquifer is present all over the basin with thickness varying from less than 10 meter in the vicinity of hilly areas to more than about 30m, in general. Occasionally, it gains more than 30m thickness, locally as seen around Ramgarh in Alwar district. In the western part of the river basin hill outcrops are present and surrounding to these outcrops; weathered and fractured zone of hard rock acts as aquifer, varying in thickness from less than 2 meter to more than 6 meter falling within <10m thickness category. West of Alwar and around Rajgarh in Alwar district, the thickness of weathered and fractured zone reaches thickness of about 20m.

Alluvial areas

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)		Total Area (sq km)
	Alwar	Bharatpur	
< 10	818.3	340.4	1,158.7
10-20	656.1	611.8	1,267.9
20-30	150.5	160.1	310.6
> 30	49.3	2.9	52.2
Total	1,674.2	1,115.2	2,789.4

Hardrock areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)		Total Area (sq km)
	Alwar	Bharatpur	
<10	262.5	-	262.5
10-20	257.3	-	257.3
> 20	2.5	-	2.5
Total	522.3	-	522.3



CROSS SECTIONS

RUPARAIL RIVER BASIN

Several hydrogeologic cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXII. The broad alignment of the sections is as given below:

Name of Section Line	Orientation
Section AA'	SW – NE
Section BB'	W – E
Section CC'	N – S
Section DD'	NW – SE

CROSS SECTIONS

RUPARAIL RIVER BASIN

Section A-A':

Section A-A' (Plate – XIX) is taken in an approximate SW-NE direction covering length of about 70km. The lithologs of 7 boreholes along with surrounding well information is taken while preparing the section. Quartzite and Slate constitute continuous bedrock over which 60-70m of alluvial material is deposited. The alluvium is formed of clay and sand. In most part of the section clay constitutes the lower layer and sand is deposited over it. A thin soil cover is persistently present over the alluvium.

Depth to water level ranges from 180 meter amsl to 215 meter amsl maintaining good parallelism to topography, except in 0-10 km stretch where it is deeper.

Section B-B':

The section BB' (Plate – XX) is considered from W to E direction covering to a distance of 45 km. The lithologs of 7 boreholes are incorporated in this section along with surrounding data. In this section too, the quartzites and slates form the bedrock base over which 80-100m thick alluvial material is deposited. The western part is predominantly sandy with small clay lens within but the eastern half of the section is primarily clayey with very limited sand thickness which however again increases further eastwards. A thin soil cover is persistently present.

Depth to water level ranges from 195 meter amsl to 225 meter amsl in the section which is deeper in western part and shallower in eastern part of the section.

PLATE - XIX

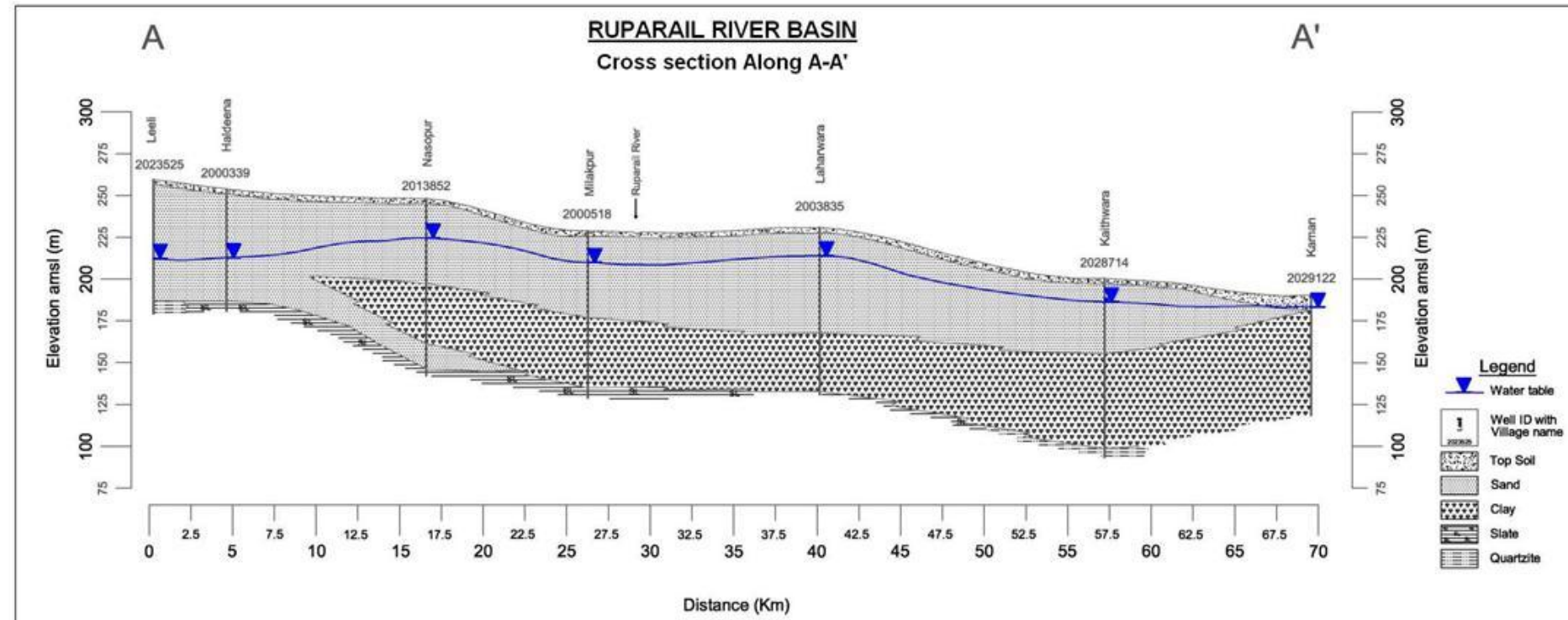
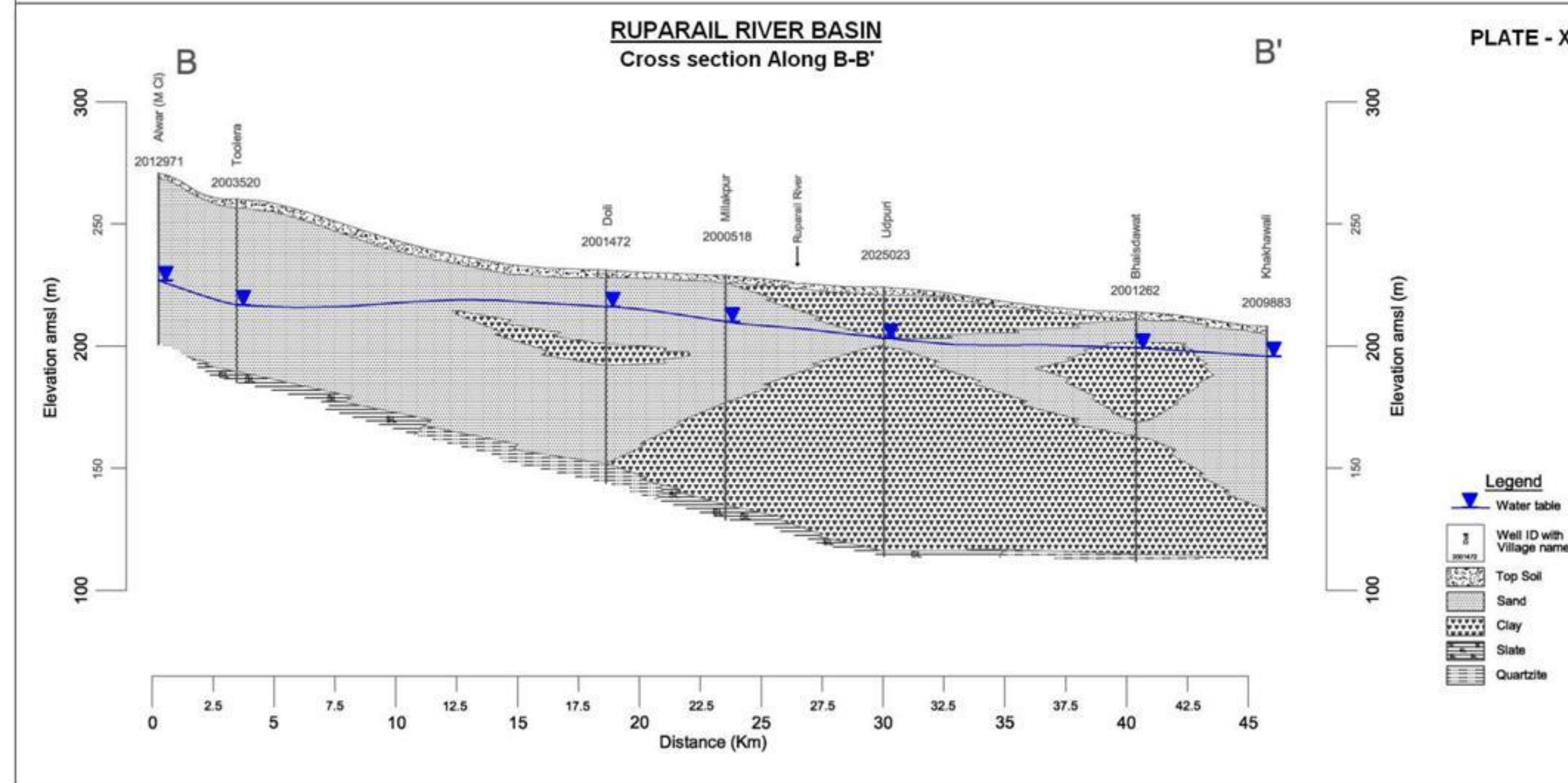


PLATE - XX



CROSS SECTIONS

RUPARAIL RIVER BASIN

Section C-C':

Section C-C' (Plate – XXI) is prepared trending N-S direction in the western part of the river basin covering a length of about 35kms. The lithologs of 5 boreholes are incorporated in the section along with surrounding data. Here also, the lowermost horizon in the section is quartzite and slaty with quartzite appearing prominently in northern part and slate in southern. The section can be described as predominantly clayey in nature with only 25-30m thick sandy beds below or above the clay layer which forms aquifers in the area. Thin soil cover constitutes the top layer.

Depth to water level ranges from 180 meter amsl to 190 meter amsl in the section.

Section D-D':

Section D-D' (Plate – XXII) is taken along NW-SE direction in the eastern part of the basin covering a length of about 16km only. The data from 4 boreholes has been utilized, with surrounding well information to decipher the sub-surface information. This section also reveals a thick horizon of alluvial material. The bedrock formed of slate occurs at a very shallow depth of about 35m below ground level which gradually steepens to about 75m bgl made up of quartzite in the next well just 3kms southwards. Quartzite again appears in the southernmost well as bedrock. The alluvial material is predominantly clayey but sand does form good aquifers in the eastern and western parts of the section.

Depth to water level ranges from 225 meter amsl to 250 meter amsl in the section and follows parallelism to topography.

PLATE - XXI

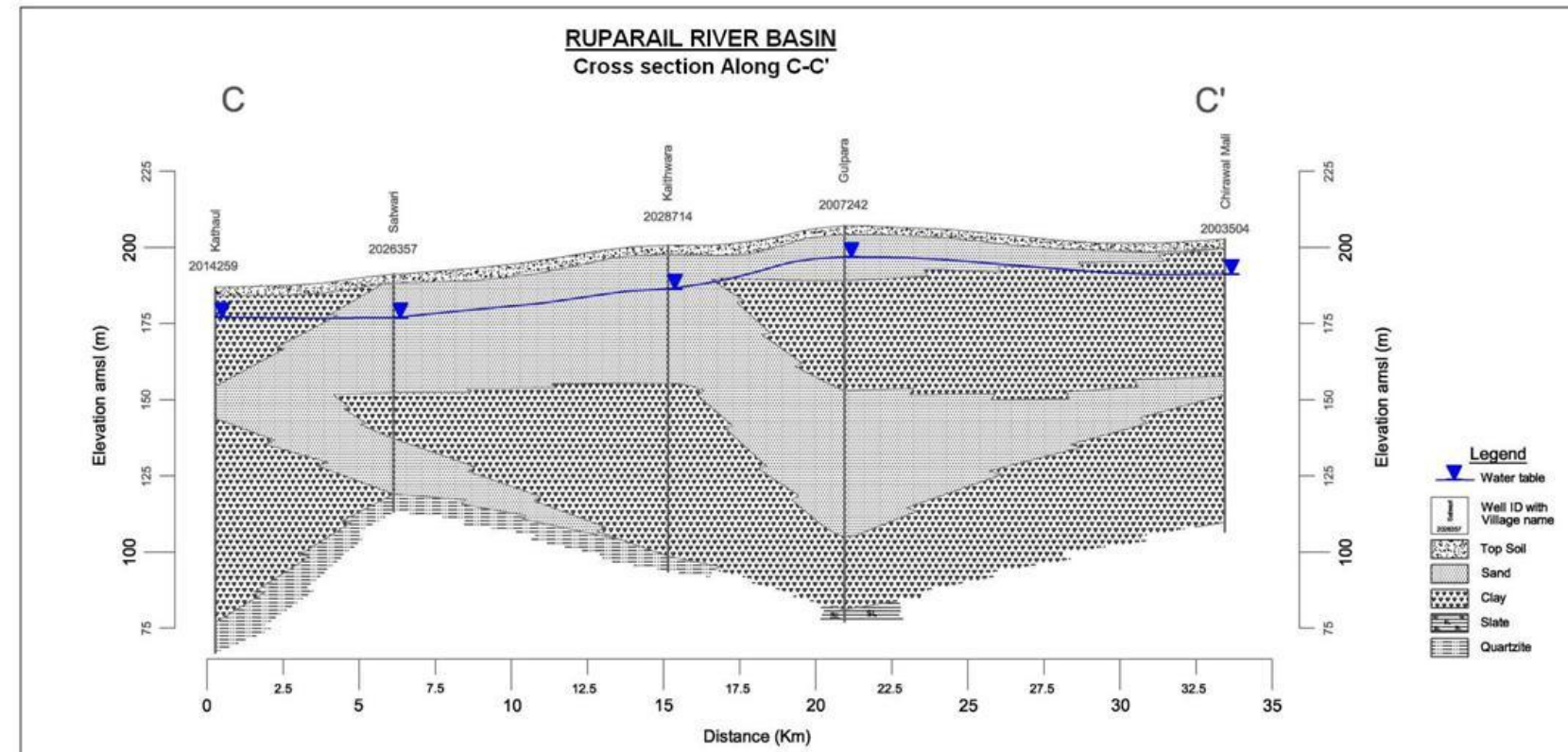
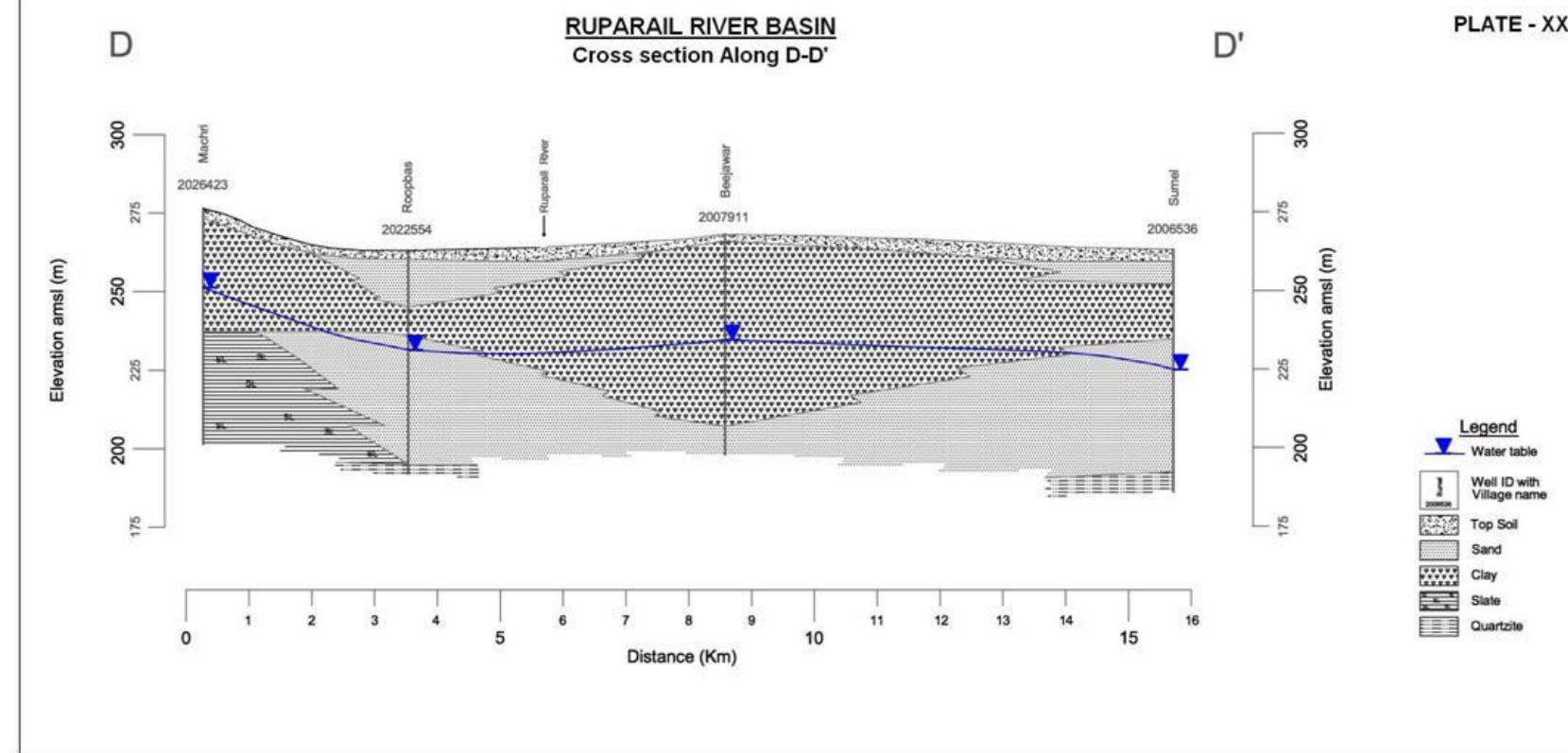


PLATE - XXII



3D MODEL OF AQUIFERS

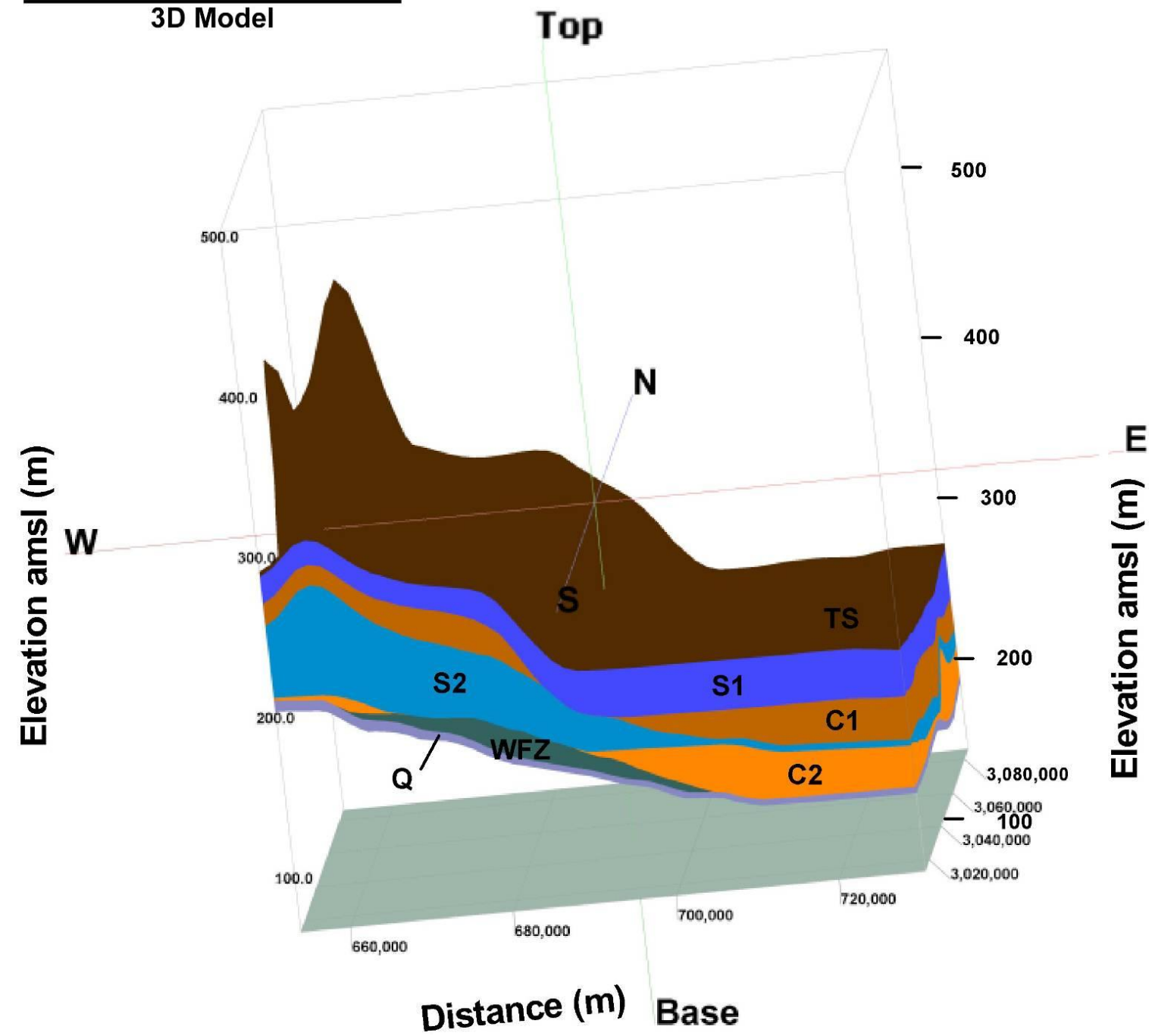
RUPARAIL RIVER BASIN

The continuous litho-stratigraphic model has been developed for the Ruparail River Basin using the data of scattered wells as input. 3D model depicts the sub-surface aquifer disposition of litho-stratigraphic units forming aquifers, aquicludes and aquitards in the area.

Plate – XXIII presents 3D model depicting the various litho-stratigraphic units in the entire river basin. From this model it is apparent that older alluvium consisting of sand and clay is underlying the top soil throughout the basin area as multiple intercalated layers. Alluvium is underlain by quartzite in the area. Significant weathered and fractured hard rock zone is present in the southern part of the area.

PLATE - XXIII

RUPARAIL RIVER BASIN
3D Model



Legend

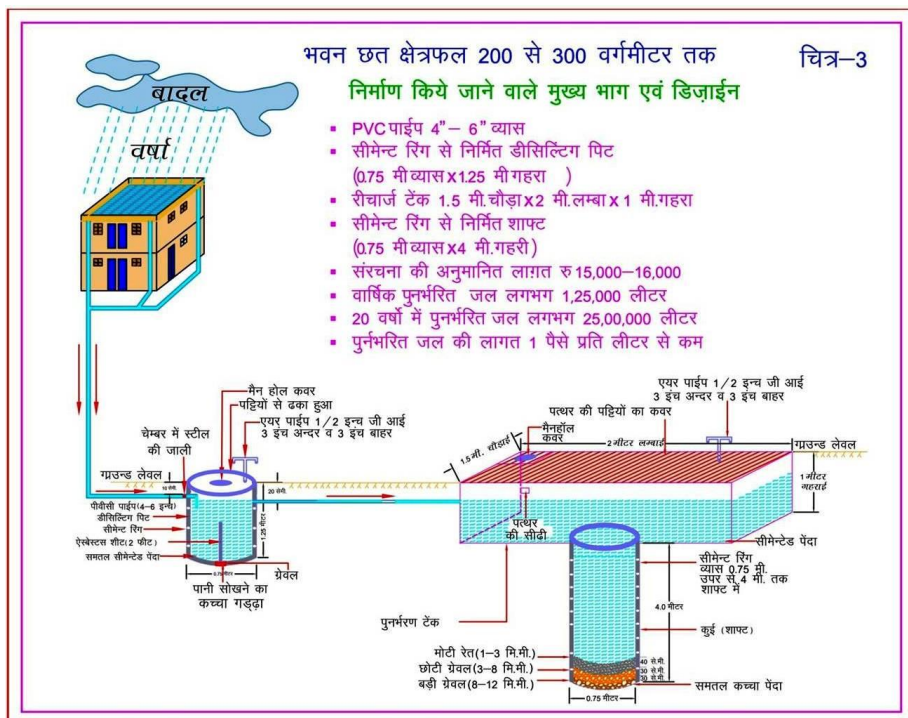
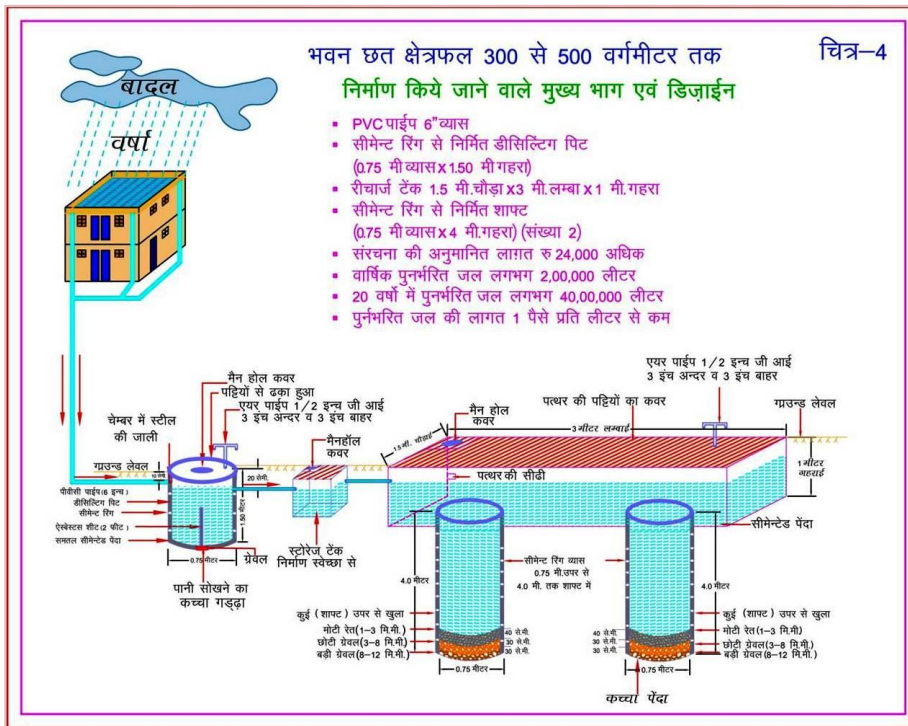
TS	Top Soil
S1	Sand
C1	Clay
S2	Sand
C2	Clay
WFZ	Weathered & Fractured Zone
Q	Quartzite

Glossary of terms

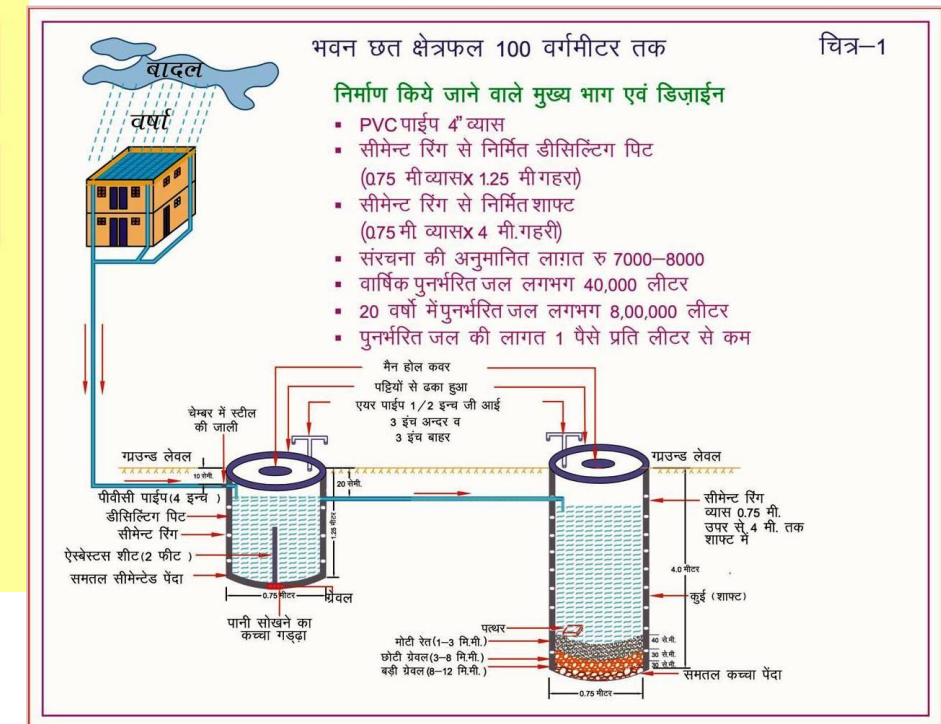
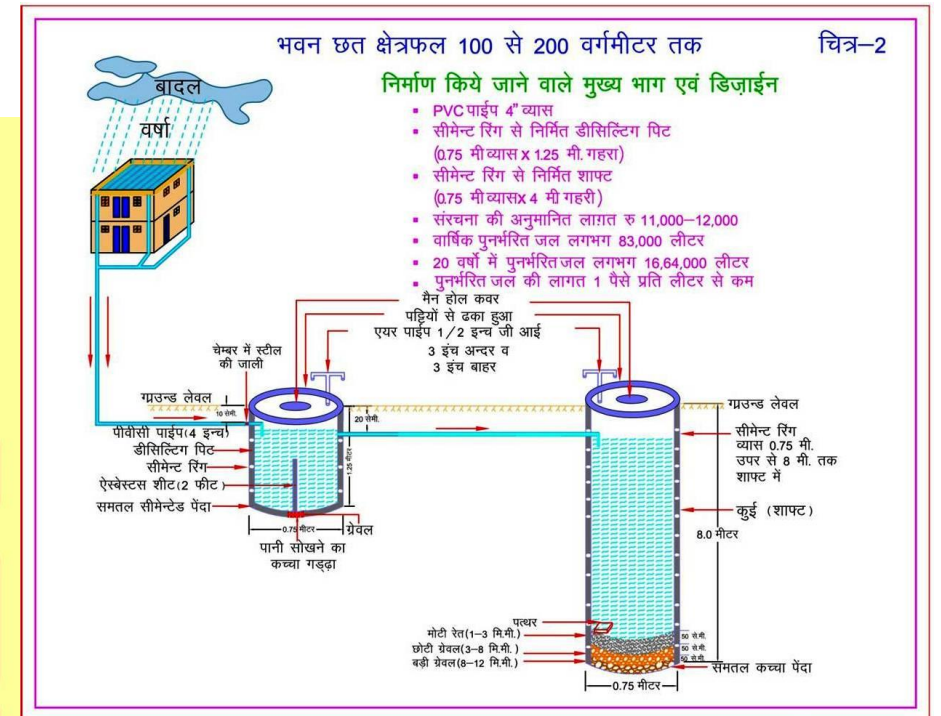
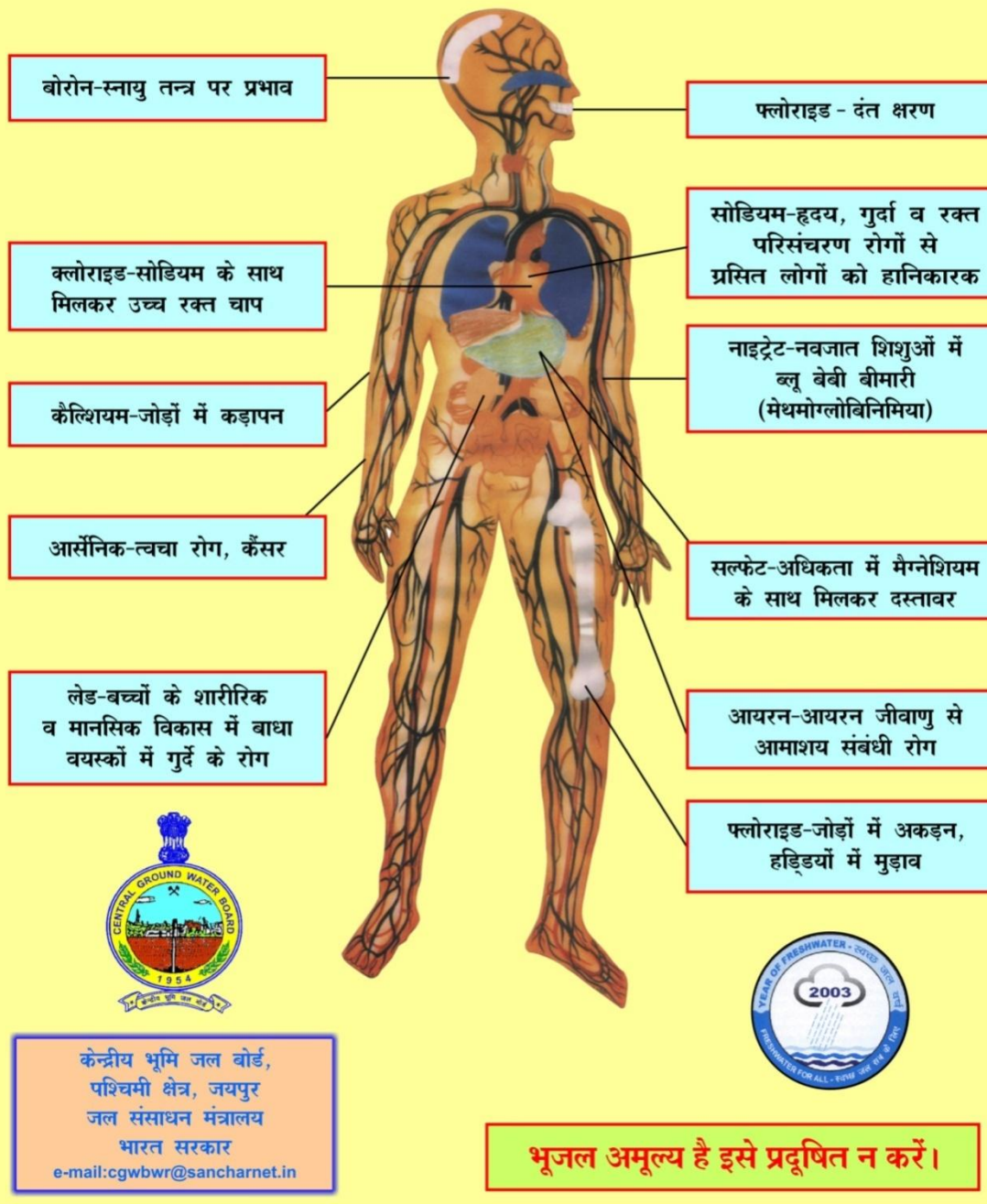
S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



ROLTA

Rolta India Limited

Central & Registered Office

Rolta Tower A,

Rolta Technology Park,

MIDC, Andheri (East),

Mumbai - 400 093

Tel : +91 (22) 2926 6666, 3087 6543

Fax : +91 (22) 2836 5992

Email : indsales@rolta.com

www.rolta.com